



Information Storage and Management



UNIT I - Introduction to Information Storage and Management

- Introduction to Information Storage and Management: Storage System Environment, Information Storage, Evolution of Storage Technology and Architecture, Data Center Infrastructure, Key Challenges in Managing Information, Information Lifecycle Components of Storage System Environment, Disk Drive Components, Disk Drive Performance, Fundamental Laws Governing Disk Performance, Logical Components of the Host, Application Requirements and Disk Performance.

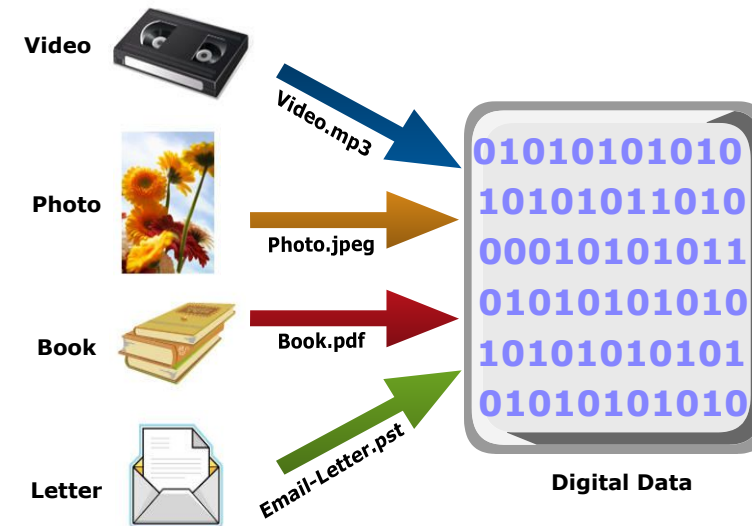
Why Information Storage

- “Digital universe – The Information Explosion”
 - 21st Century is information era
 - Information is being created at ever increasing rate
 - Information has become critical for success
- We live in an on-command, on-demand world
 - Example: Social networking sites, e-mails, video and photo sharing website, online shopping, search engines etc
- Information management is a **big** challenge
 - Organization seek to *Store* → *Protect* → *Optimize* → *Leverage* the information optimally

What is Data

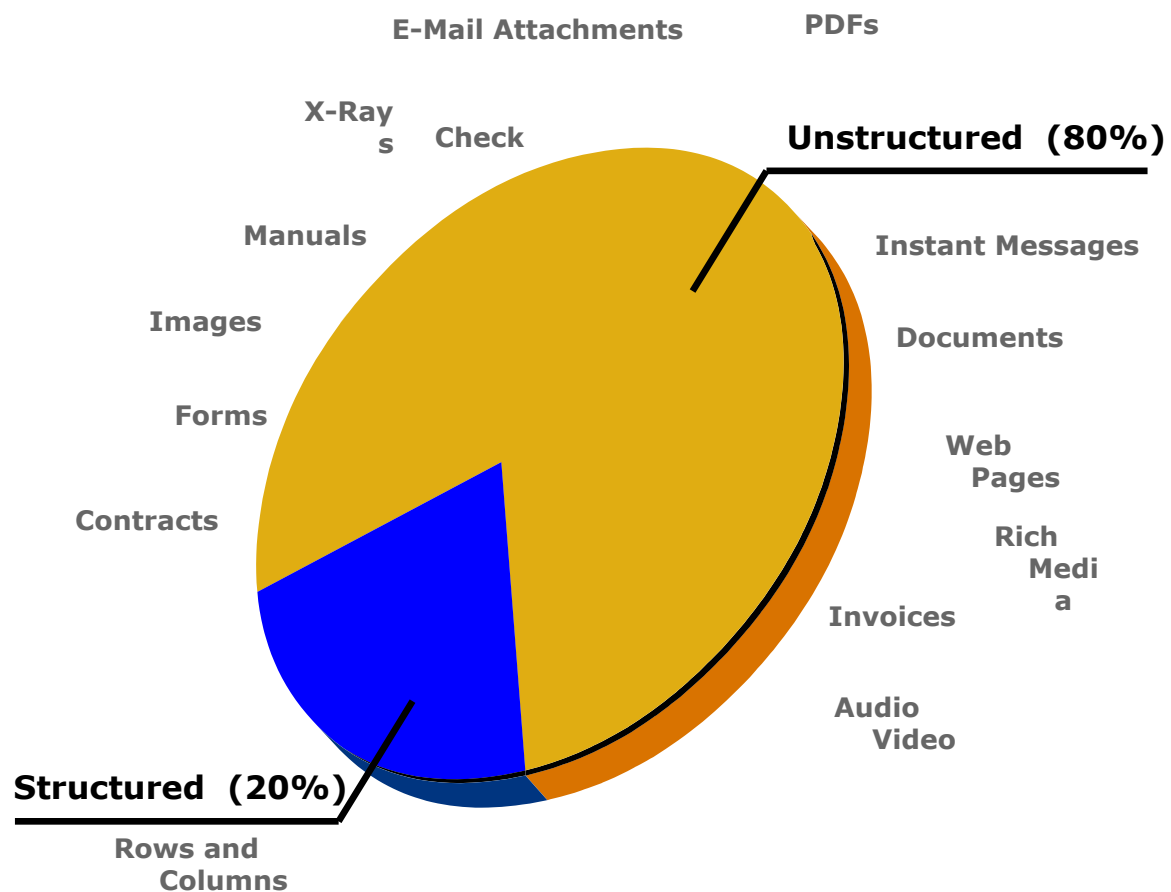
“Collection of raw facts from which conclusions may be drawn”

- Data is converted into more convenient form i.e. Digital Data
 - Increase in data processing capabilities
 - Lower cost of digital storage
 - Affordable and faster communication technology
- Who creates data?
 - Individuals
 - Businesses



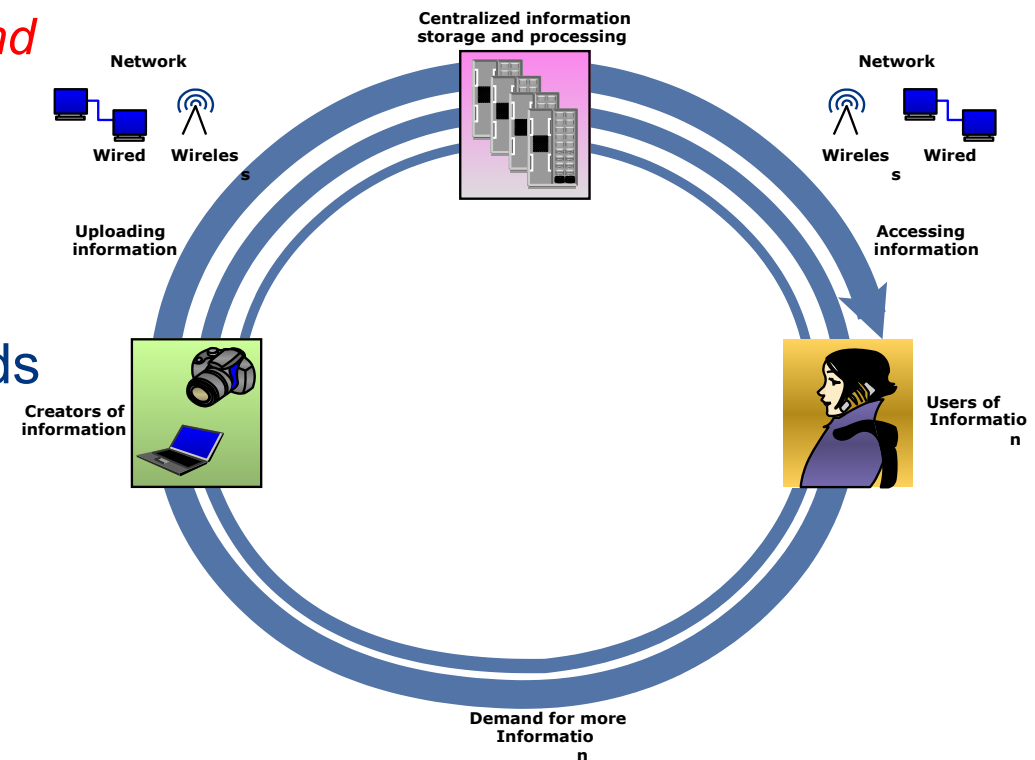
Categories of Data

- Data can be categorized as either structured or unstructured data
- Over 80% of enterprise information is unstructured



Define Information

- What do individuals/businesses do with the data they collect?
 - They turn it into “information”
 - *“Information is the intelligence and knowledge derived from data”*
- Businesses analyze raw data in order to identify meaningful trends
 - For example:
 - Buying habits and patterns of customers
 - Health history of patients



Virtuous cycle of information

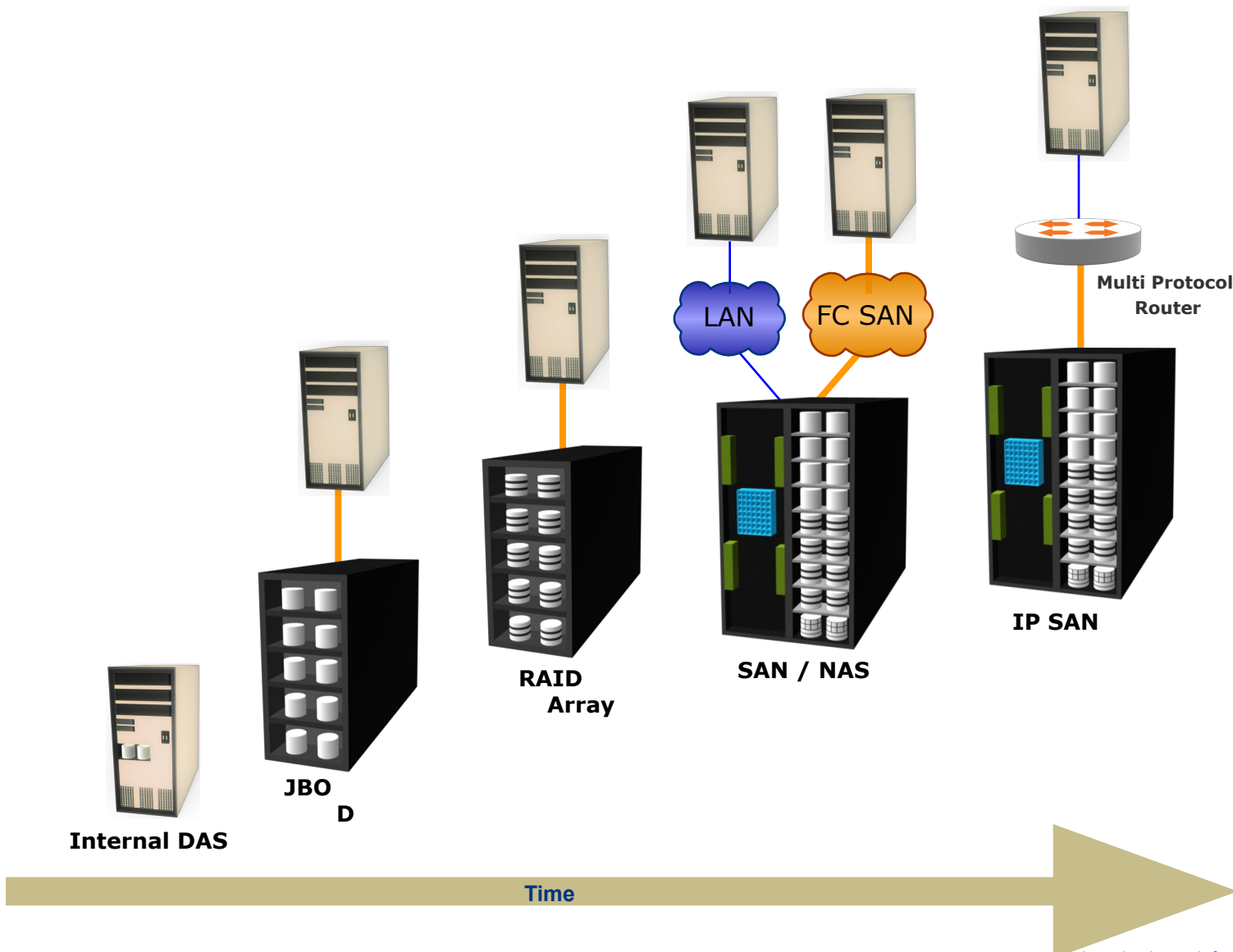
Value of Information to a Business

- Identifying new business opportunities
 - Buying/spending patterns
 - Internet stores, retail stores, supermarkets
 - Customer satisfaction/service
 - Tracking shipments, and deliveries
- Identifying patterns that lead to changes in existing business
 - Reduced cost
 - Just-in-time inventory, eliminating over-stocking of products, optimizing shipment and delivery
 - New services
 - Security alerts for “stolen” credit card purchases
 - Targeted marketing campaigns
 - Communicate to bank customers with high account balances about a special savings plan
- Creating a competitive advantage

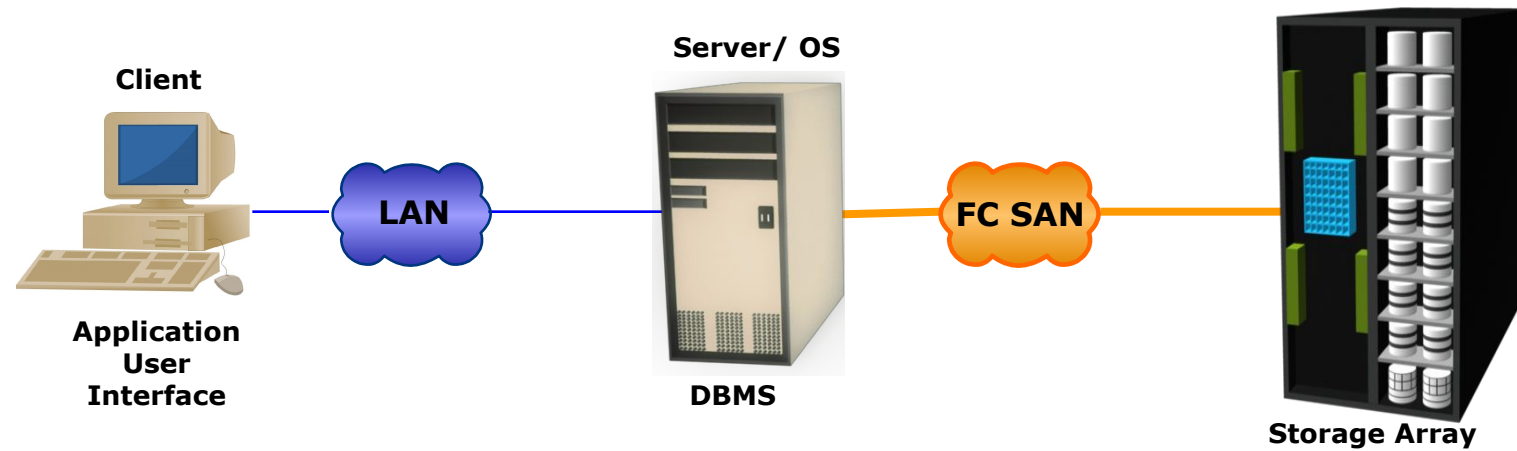
Storage

- Data created by individuals/businesses must be stored for further processing
- Type of storage used is based on the type of data and the rate at which it is created and used
- Examples:
 - Individuals: Digital camera, Cell phone, DVD's, Hard disk
 - Businesses: Hard disk, external disk arrays, tape library
- Storage model: An evolution
 - Centralized: mainframe computers
 - Decentralized: Client –server model
 - Centralized: Storage Networking

Storage Technology and Architecture Evolution



Example of an Order Processing System



Challenges in Managing Information

- **Exploding digital universe**
 - Multifold increase of information growth
- **Increasing dependency on information**
 - The strategic use of information
- **Changing value of information**
 - Information that is valuable today may become less important tomorrow.

Components of a Storage System Environment

The three main components in a storage system environment the host, connectivity, and storage are described in this section.

Host

Users store and retrieve data through applications. The computers on which these applications run are referred to as hosts.

Hosts can range from simple laptops to complex clusters of servers.

A host consists of physical components (hardware devices) that communicate with one another using logical components (software and protocols).

Access to data and the overall performance of the storage system environment depend on both the physical and logical components of a host .

Host

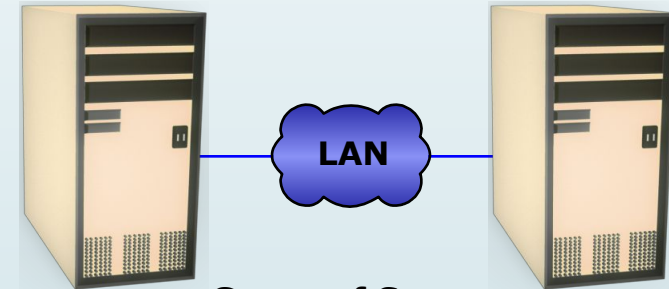
- Applications runs on hosts
- Hosts can range from simple laptops to complex server clusters
- Physical components of host
 - CPU
 - Storage
 - Disk device and internal memory
 - I/O device
 - Host to host communications
 - Network Interface Card (NIC)
 - Host to storage device communications
 - Host Bus Adapter (HBA)



Laptop



Server

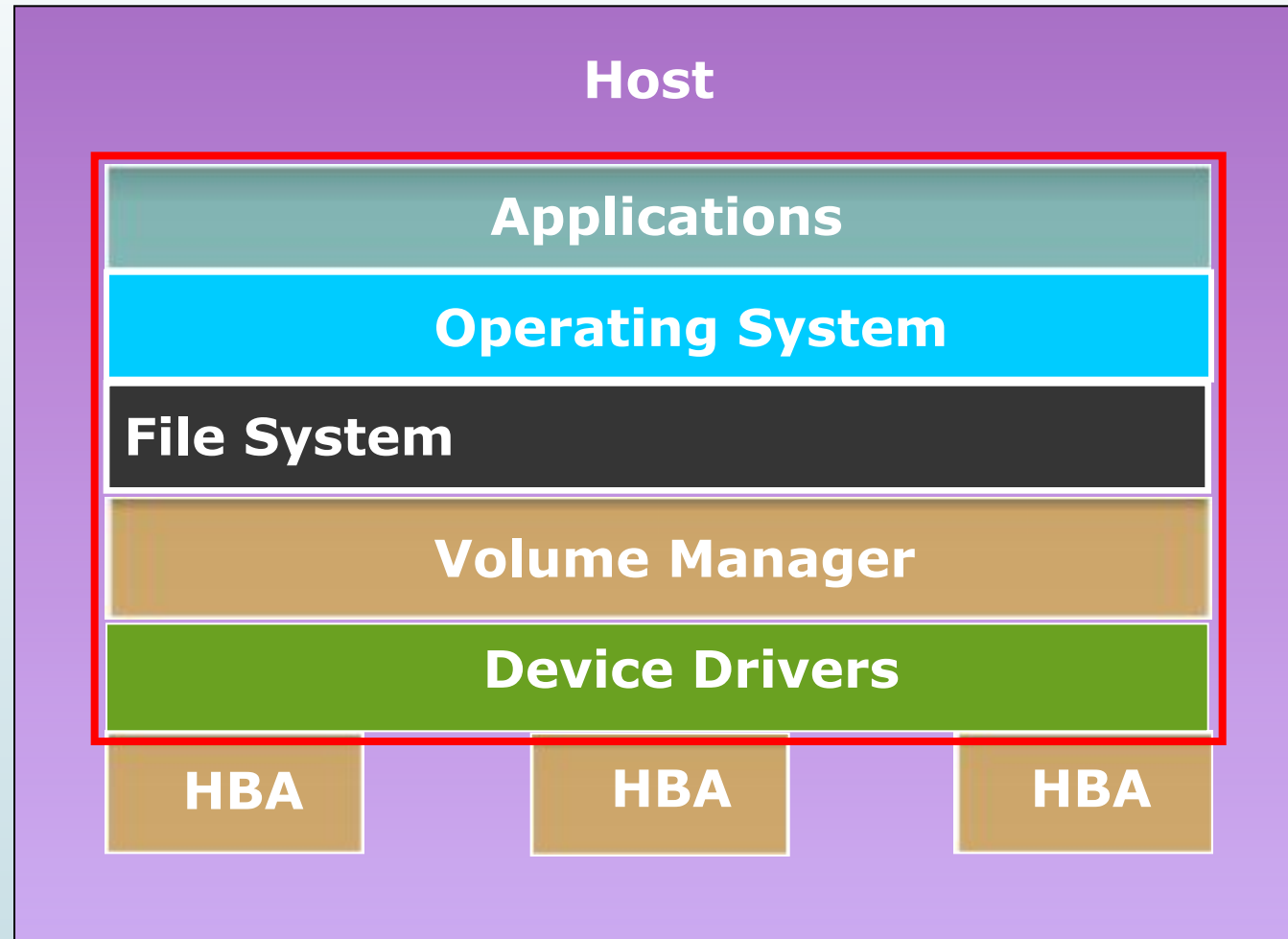


Group of Servers



Mainframe

Host: Logical Components



Logical Components of the Host

□ *Application*

- Interface between user and the host
- Three-tiered architecture
 - Application UI, computing logic and underlying databases
- Application data access can be classified as:
 - **Block-level access:** Data stored and retrieved in blocks, specifying the LBA
 - **File-level access:** Data stored and retrieved by specifying the name and path of files

□ Operating system

- Resides between the applications and the hardware
- Controls the environment

Logical Components of the Host (Cont)

- Device Drivers

- Enables operating system to recognize the device
- Provides API to access and control devices
- Hardware dependent and operating system specific

- File System

- File is a collection of related records or data stored as a unit
- File system is hierarchical structure of files
 - Examples: FAT 32, NTFS, UNIX FS and EXT2/3

File System: Metadata Examples

UNIX (UFS)

- File type and permissions
- Number of links
- Owner and group IDs
- Number of bytes in the file
- Last file access
- Last file modification

Windows (NTFS)

- Time stamp and link count
- File name
- Access rights
- File data
- Index information
- Volume information

File Systems: Journaling and Logging

- Improves data integrity and system restart time over non-journaling file systems.
- Uses a separate area called a log or journal.
 - May hold all data to be written
 - May hold only metadata
- Disadvantage - slower than other file systems.
 - Each file system update requires at least 1 extra write – to the log

Connectivity

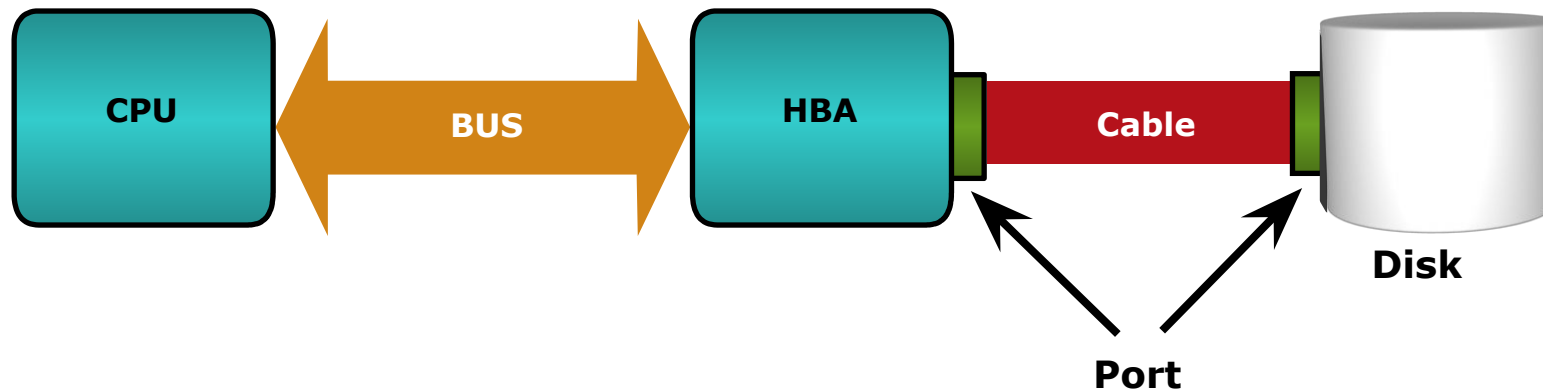
The discussion here focuses on the connectivity between the host and the storage device.

The components of connectivity in a storage system environment can be classified as **Physical** and **Logical**.

The physical components are the hardware elements that connect the host to storage and the logical components of connectivity are the protocols used for communication between the host and storage.

Physical Components of Connectivity

- Interconnection between hosts or between a host and any storage devices
- Physical Components of Connectivity are:
 - Bus, port and cable



The bus is the collection of paths that facilitates data transmission from one part of a computer to another, such as from the CPU to the memory.

The port is a specialized outlet that enables connectivity between the host and external devices.

Cables connect hosts to internal or external devices using copper or fiber optic media.

Physical components communicate across a bus by sending bits (control, data, and address) of data between devices.

These bits are transmitted through the bus in either of the following ways:

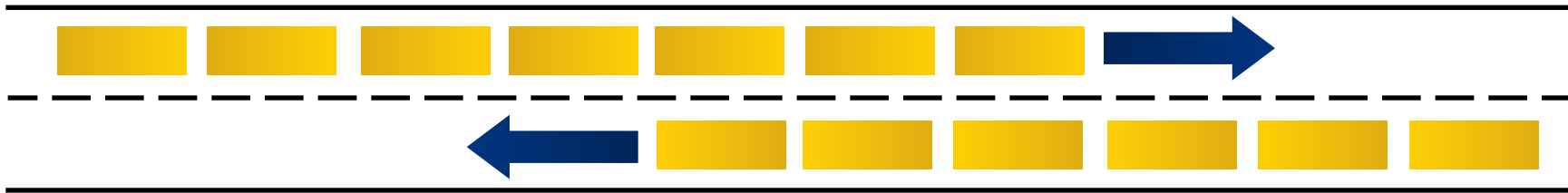
- **Serially**
- **In parallel**

Bus Technology

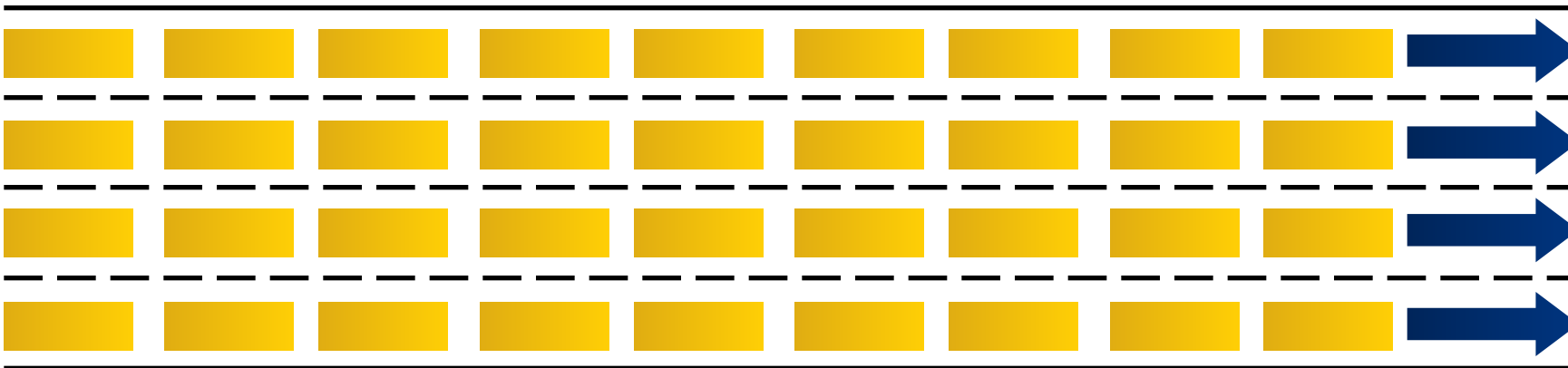
Serial



Serial Bi-directional



Parallel



Bus Technology

- System Bus – connects CPU to Memory
- Local (I/O) Bus – carries data to/from peripheral devices.
- Bus width measured in bits
- Bus speed measured in MHz
- Throughput measured in MB/S

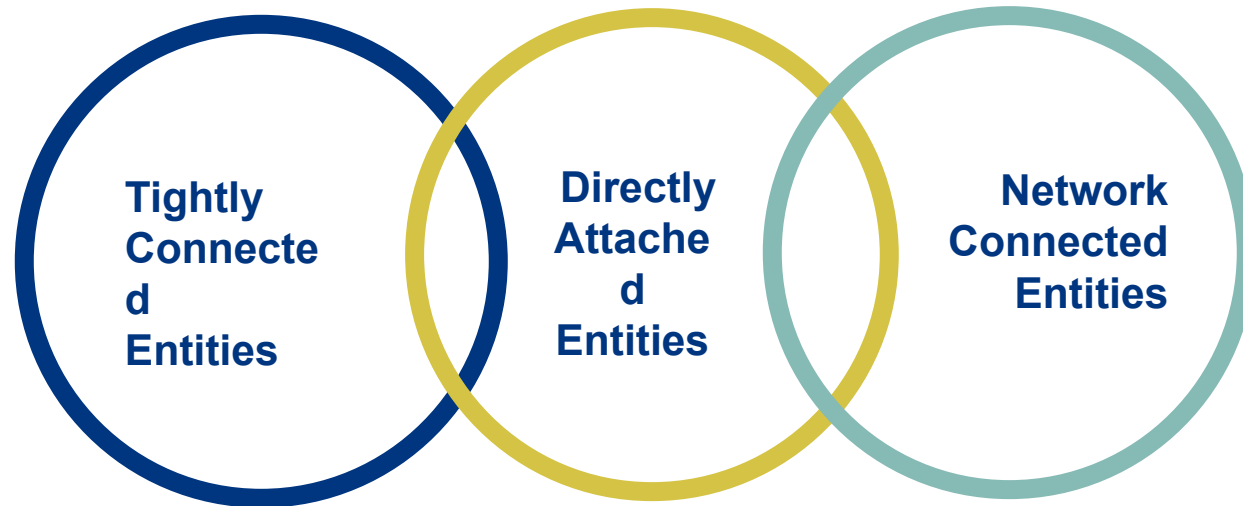
Logical Components of Connectivity

The popular interface protocol used for the local bus to connect to a peripheral device is peripheral component interconnect (PCI). The interface protocols that connect to disk systems are Integrated Device Electronics/Advanced Technology Attachment (IDE/ATA) and Small Computer System Interface (SCSI).

- **PCI**
- **IDE/ATA**
- **SCSI**

Connectivity Protocol

- Protocol = a defined format for communication between sending and receiving devices

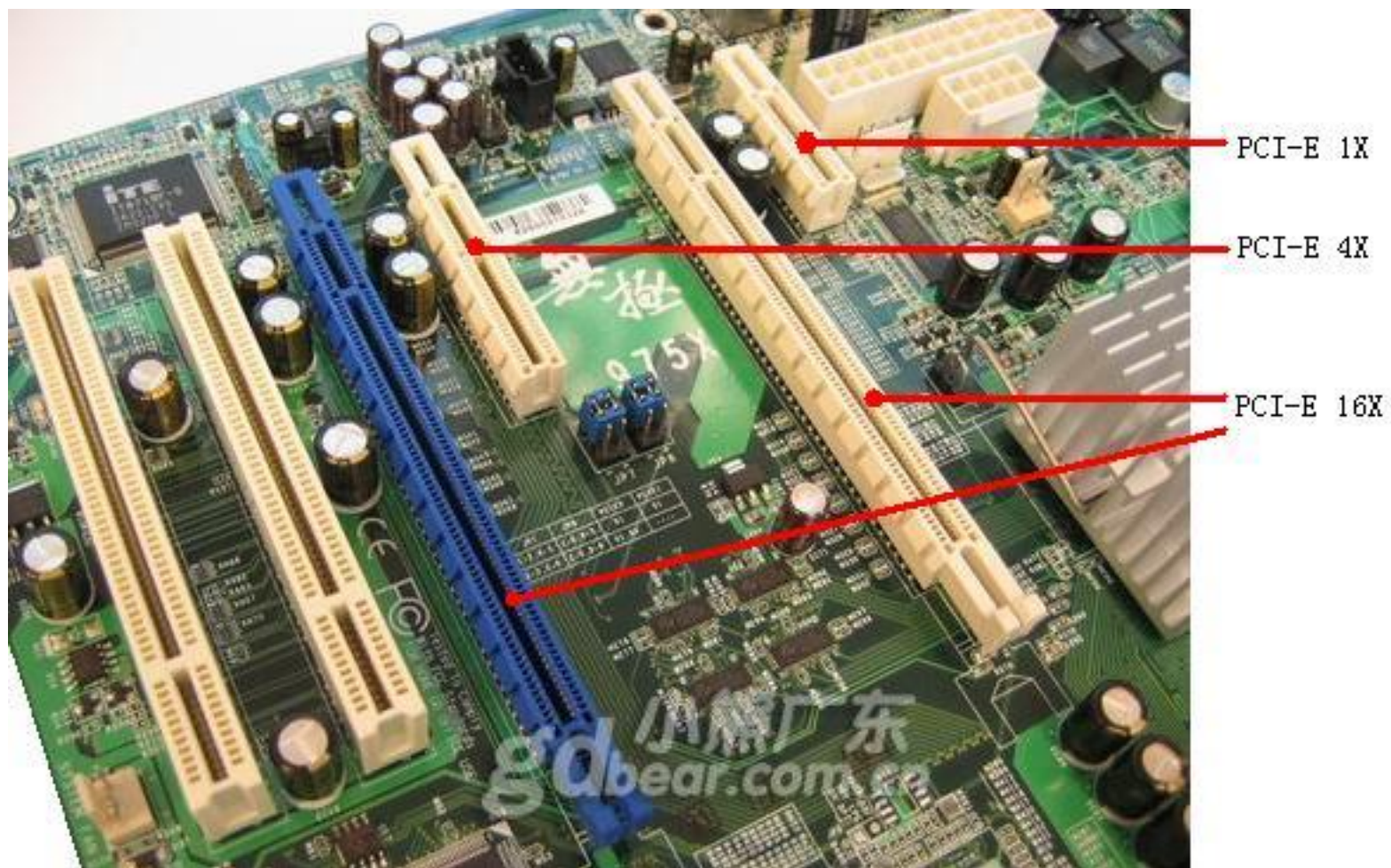


- Tightly connected entities such as central processor to RAM, or storage buffers to controllers (example PCI)
- Directly attached entities connected at moderate distances such as host to storage (example IDE/ATA)
- Network connected entities such as networked hosts, NAS or SAN (example SCSI or FC)

Popular Connectivity Options: PCI

- PCI is used for local bus system within a computer
- It is an interconnection between microprocessor and attached devices
- Has Plug and Play functionality
- PCI is 32/64 bit
- Throughput is 133 MB/sec
- *PCI Express*
 - Enhanced version of PCI bus with higher throughput and clock speed

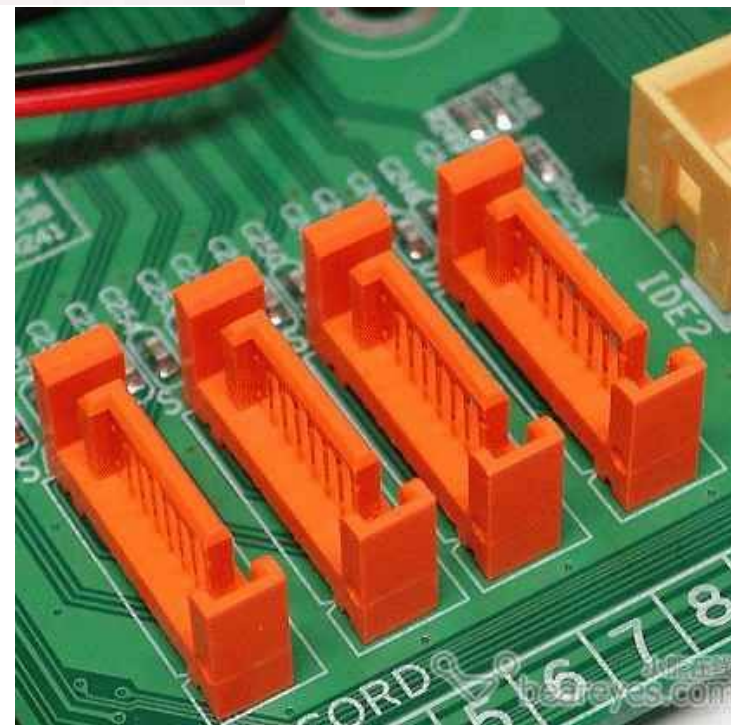
PCI



Popular Connectivity Options: IDE/ATA

- Integrated Device Electronics (IDE) / Advanced Technology Attachment (ATA)
 - Most popular interface used with modern hard disks
 - Good performance at low cost
 - Inexpensive storage interconnect
 - Used for internal connectivity
- Serial Advanced Technology Attachment (SATA)
 - Serial version of the IDE /ATA specification
 - Hot-pluggable
 - Enhanced version of bus provides upto 6Gb/s (revision 3.0)

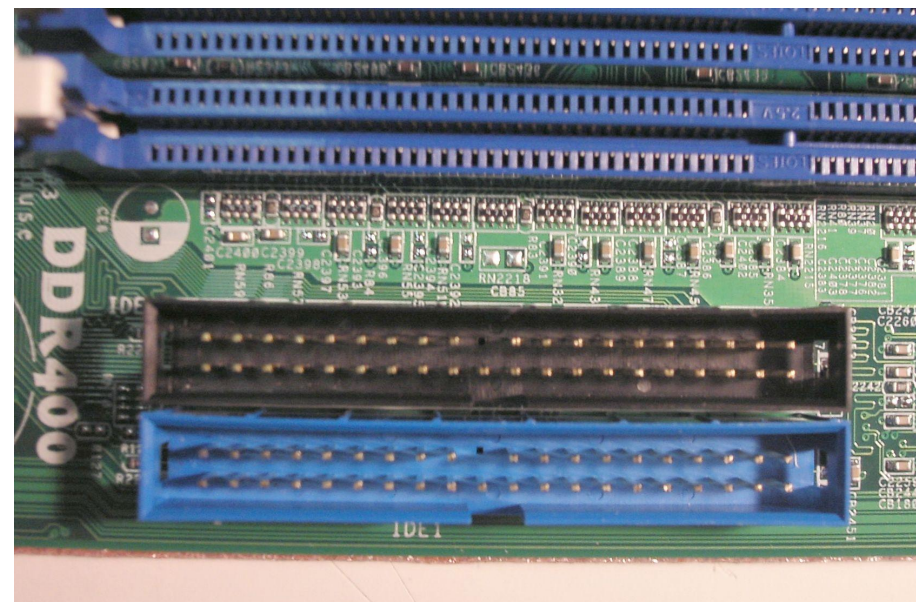
SATA



PATA



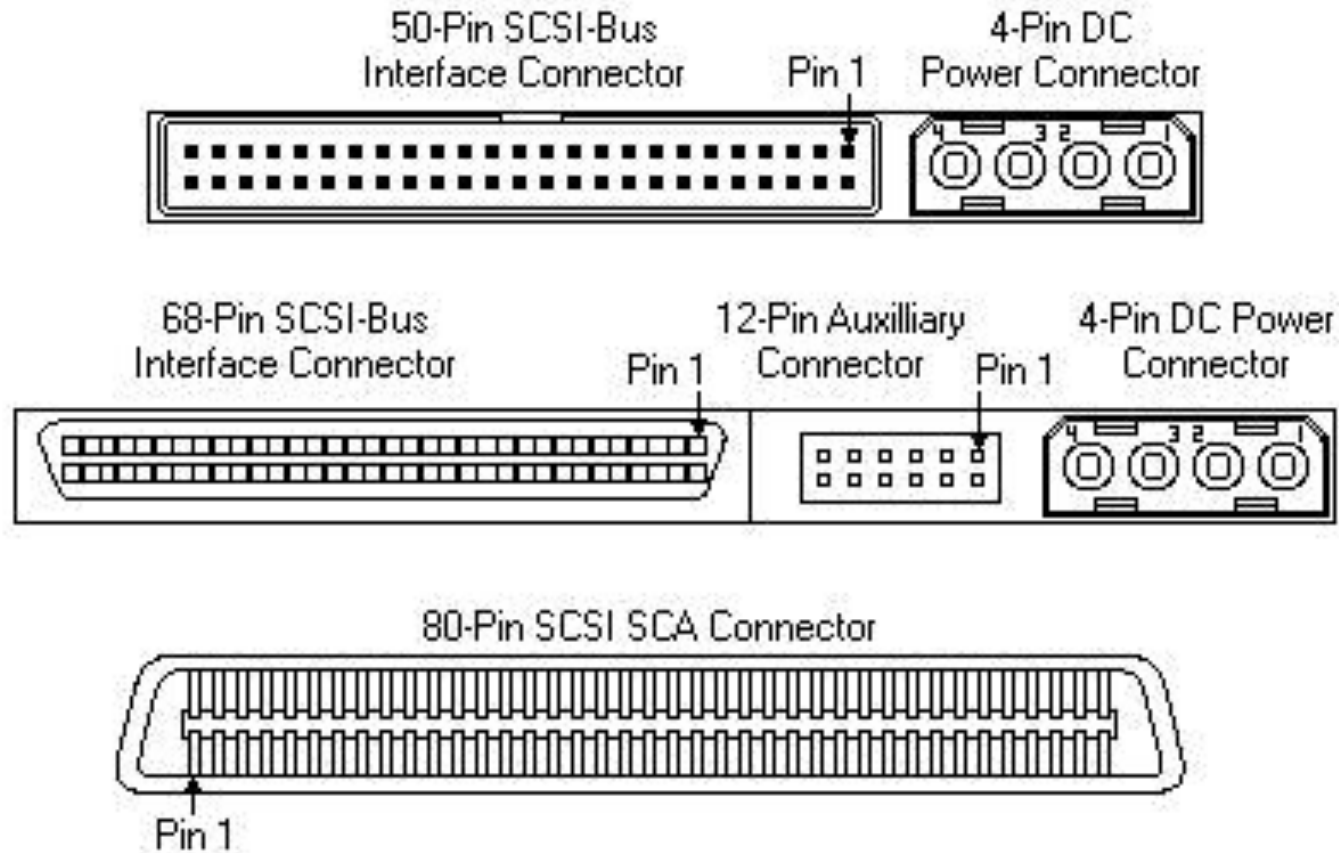
www.ChinaVisual.com 视觉中国



Popular Connectivity Options: SCSI

- **Parallel SCSI (Small computer system interface)**
 - Most popular hard disk interface for servers
 - Higher cost than IDE/ATA
 - Supports multiple simultaneous data access
 - Used primarily in “higher end” environments
 - SCSI Ultra provides data transfer speeds of 320 MB/s
- **Serial SCSI**
 - Supports data transfer rate of 3 Gb/s (SAS 300)

SCSI-Bus Interface Connector



SCSI - Pros and Cons

- Pros:

- Fast transfer speeds, up to 320 megabytes per second
- Reliable, durable components
- Can connect many devices with a single bus, more than just HDs
- SCSI host cards can be put in almost any system
- Full backwards compatibility

- Cons:

- Configuration and setup specific to one computer
- Unlike IDE, few BIOS support the standard
- Overwhelming number of variations in the standard, hardware, and connectors
- No common software interfaces and protocol

Comparison IDE/ATA vs SCSI

Feature	IDE/ATA	SCSI
Connectivity Market	Internal Storage	Internal and External Storage
Speed (MB/sec)	100/133/150	320
Hot Pluggable	No	Yes
Expandability	Easier to set up	Very good but very expensive to set up
Cost/Performance	Good	High cost/Fast transfer speed

Storage: Medias and Options

- ❑ Magnetic Tape
 - ❑ Low cost solution for long term data storage
 - ❑ Limitations
 - ❑ Sequential data access, Single application access at a time, Physical wear and tear and Storage/retrieval overheads
- ❑ Optical Disks
 - ❑ Popularly used as distribution medium in small, single-user computing environments
 - ❑ Write once and read many (WORM): CD-ROM, DVD-ROM
 - ❑ Limited in capacity and speed
- ❑ Disk Drive
 - ❑ Most popular storage medium with large storage capacity
 - ❑ Random read/write access
 - ❑ Ideal for performance intensive online application

Storage: Medias and Options

- Magnetic Tape
 - Low cost solution for long term data storage
 - Limitations
 - Sequential data access, Single application access at a time, Physical wear and tear and Storage/retrieval overheads
- Optical Disks
 - Popularly used as distribution medium in small, single-user computing environments
 - Write once and read many (WORM): CD-ROM, DVD-ROM
 - Limited in capacity and speed
- Disk Drive
 - Most popular storage medium with large storage capacity
 - Random read/write access
 - Ideal for performance intensive online application

Disk Drive Components

A disk drive uses a rapidly moving arm to read and write data across a flat platter coated with magnetic particles.

Data is transferred from the magnetic platter through the R/W head to the computer.

Several platters are assembled together with the R/W head and controller, most commonly referred to as a hard disk drive (HDD).

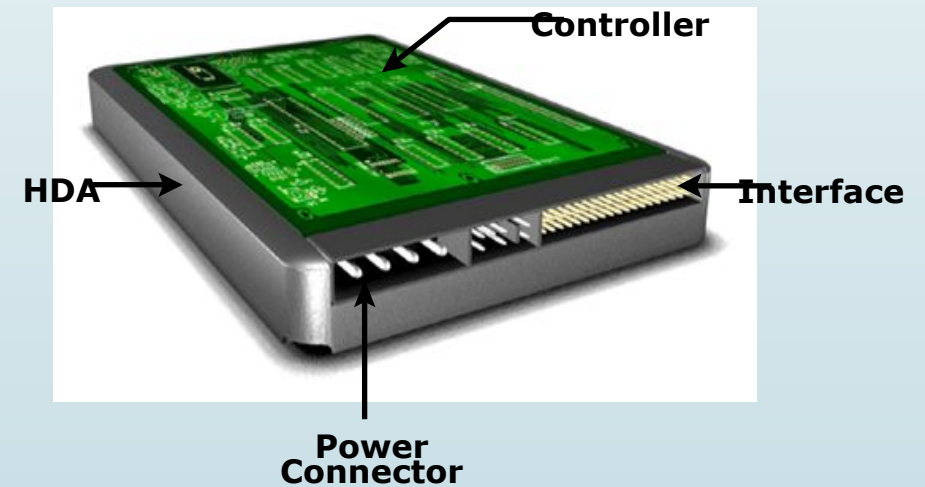
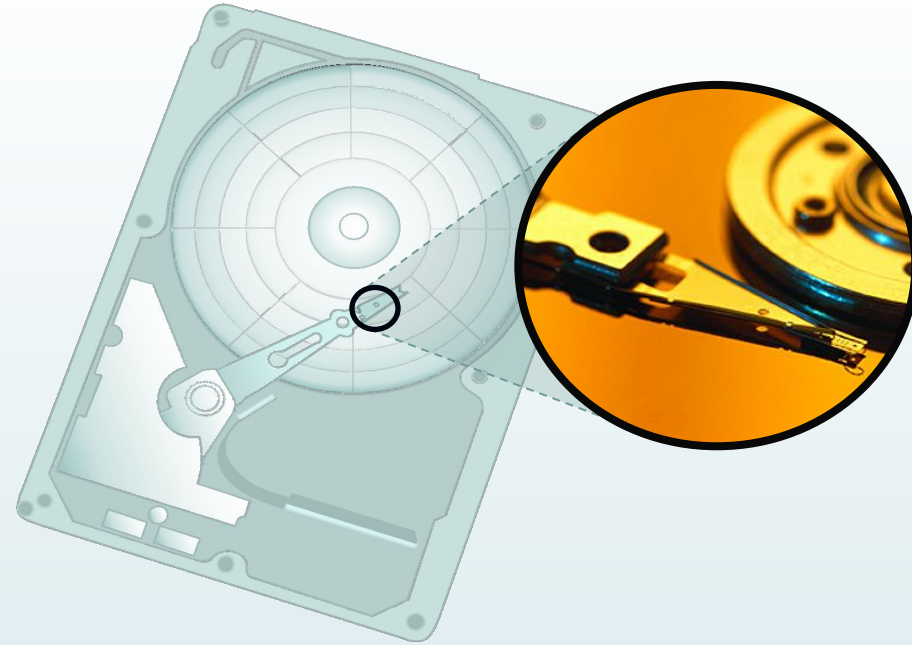
Data can be recorded and erased on a magnetic disk any number of times.

This section details the different components of the disk, the mechanism for organizing and storing data on disks, and the factors that affect disk performance.

Disk Drive Components

Key components of a disk drive:

- **Platter**
- **Spindle**
- **Read/Write Head**
- **Actuator Arm Assembly**
- **Controller**
- **Physical Disk Structure**
- **Zoned Bit Recording**
- **Logical Block Addressing**



Platter

40

A typical HDD consists of one or more flat circular disks called platters.

The data is recorded on these platters in binary codes (0s and 1s).

The set of rotating platters is sealed in a case, called a Head Disk Assembly (HDA).

A platter is a rigid, round disk coated with magnetic material on both surfaces (top and bottom).

The data is encoded by polarizing the magnetic area, or domains, of the disk surface.

Spindle

A spindle connects all the platters, as shown in Figure 2-3, and is connected to a motor.

The motor of the spindle rotates with a constant speed. The disk platter spins at a speed of several thousands of revolutions per minute (rpm).

Disk drives have spindle speeds of 7,200 rpm, 10,000 rpm, or 15,000 rpm.

Disks used on current storage systems have a platter diameter of 3.5" (90 mm).

When the platter spins at 15,000 rpm, the outer edge is moving at around 25 percent of the speed of sound.

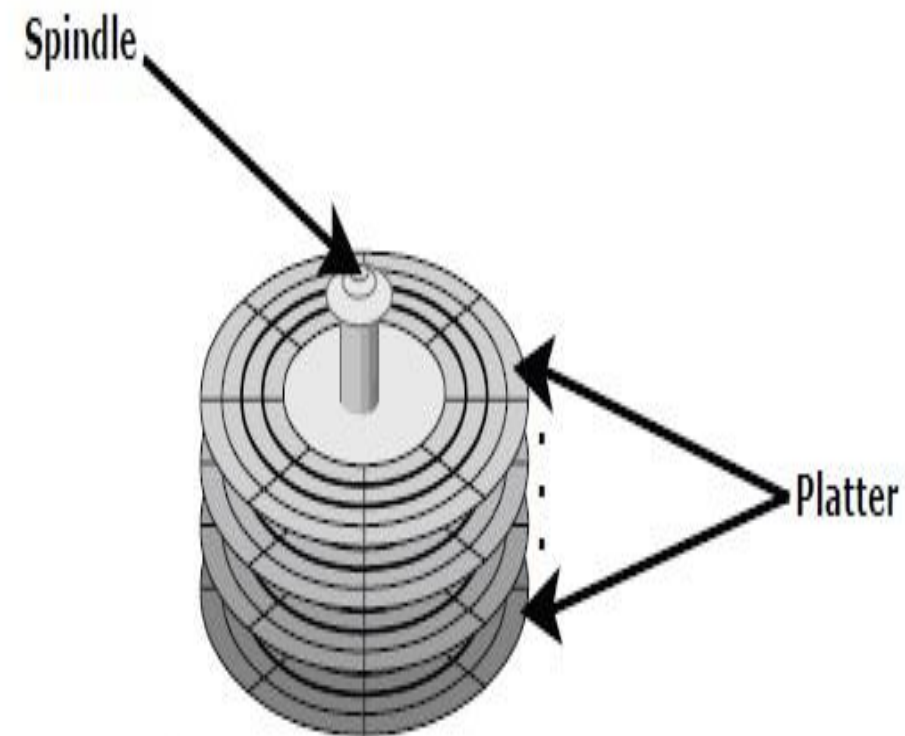


Figure 2-3: Spindle and platter

Physical Disk Structure

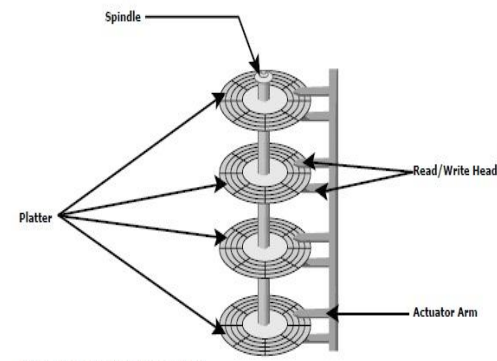
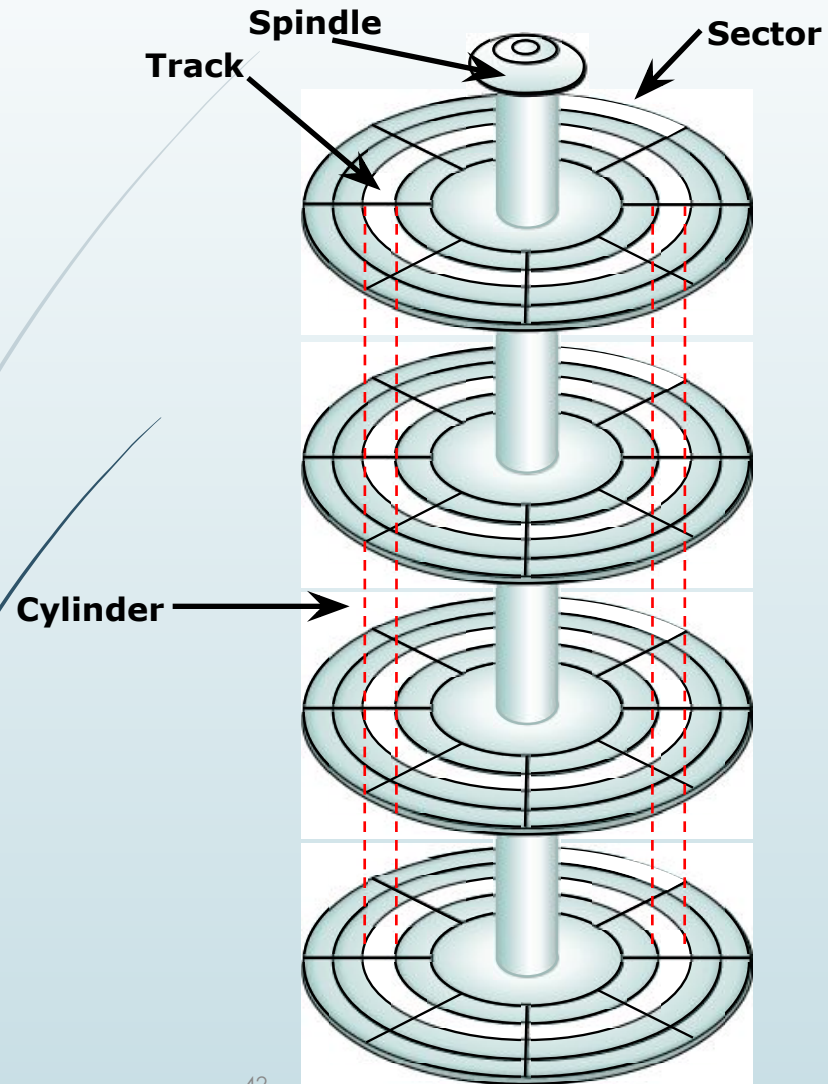
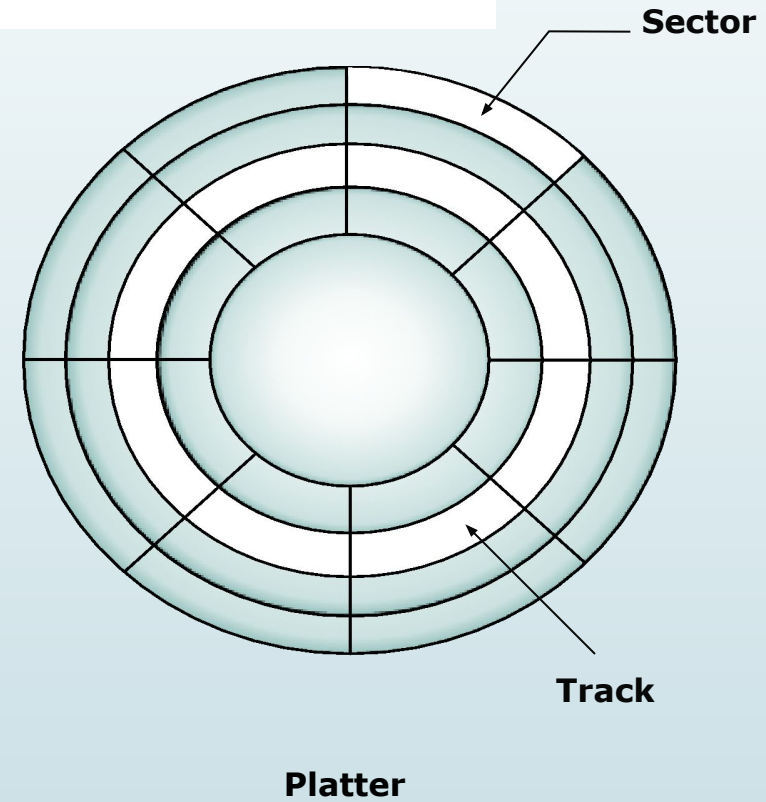
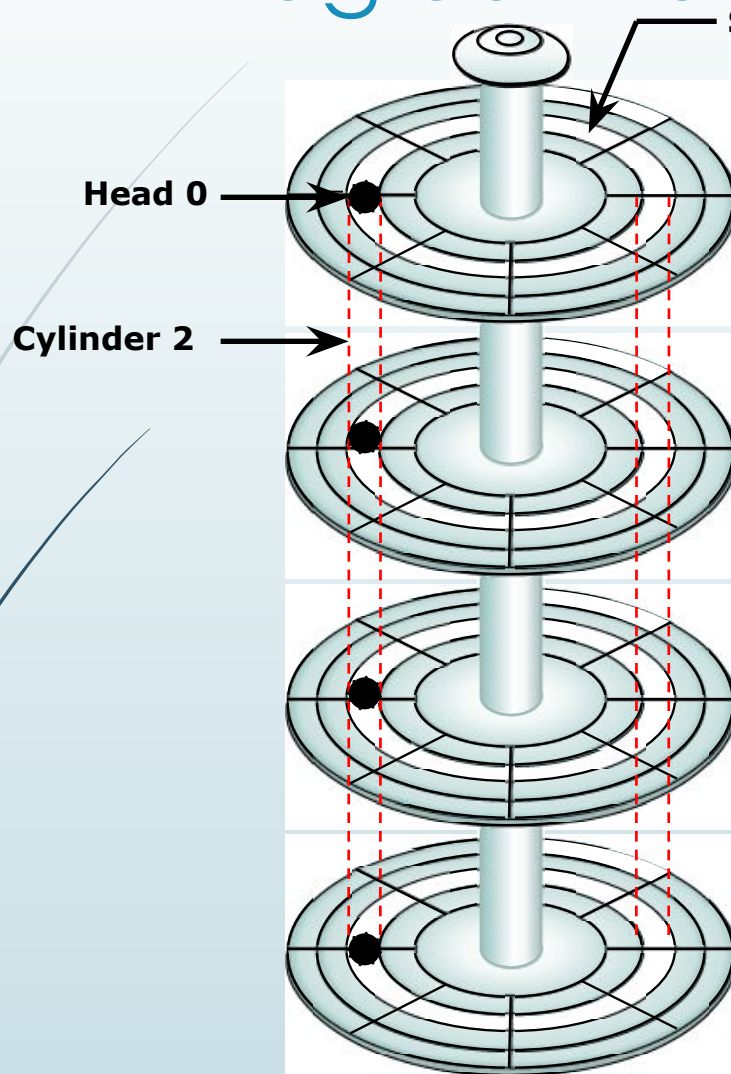


Figure 2-4: Actuator arm assembly

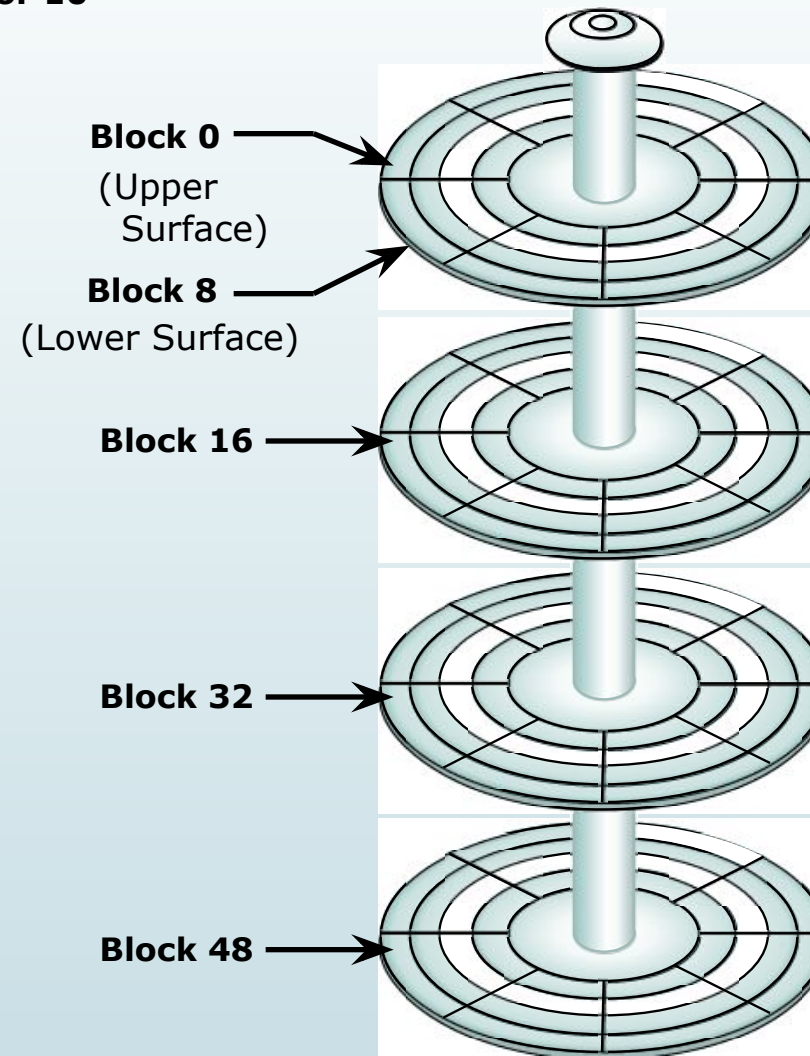


Logical Block Addressing



- 43

Physical Address= CHS



Logical Block Address= Block#

Zoned Bit Recording

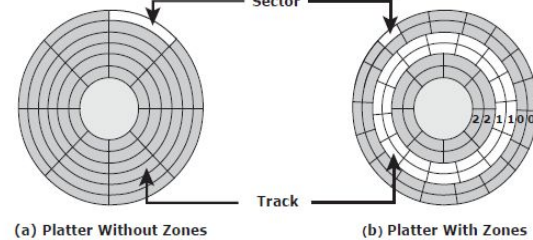
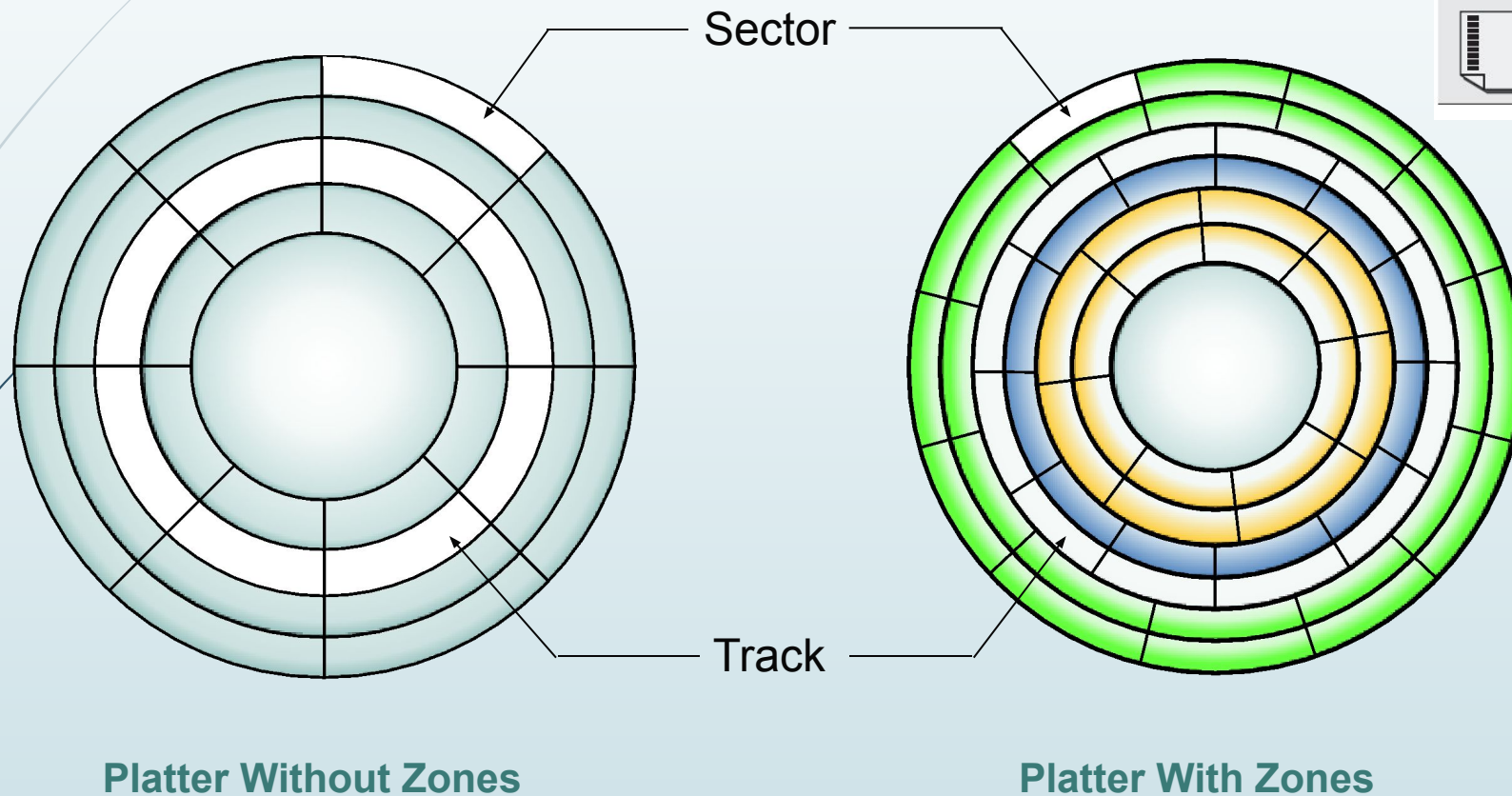
Because the platters are made of concentric tracks, the outer tracks can hold more data than the inner tracks, because the outer tracks are physically longer than the inner tracks, as shown in Figure 2-6 (a).

On older disk drives, the outer tracks had the same number of sectors as the inner tracks, so data density was low on the outer tracks.

This was an inefficient use of available space. Zone bit recording utilizes the disk efficiently.

As shown in Figure 2-9 (b), this mechanism groups tracks into zones based on their distance from the center of the disk. The zones are numbered, with the outermost zone being zone 0.

Platter Geometry and Zoned-Bit Recording



The data transfer rate drops while accessing data from zones closer to the center of the platter. Applications that demand high performance should have their data on the outer zones of the platter.

Read/Write Head

Read/Write (R/W) heads, shown in Figure 2-4, read and write data from or to a platter.

Drives have two R/W heads per platter, one for each surface of the platter.

The R/W head changes the magnetic polarization on the surface of the platter when writing data.

While reading data, this head detects magnetic polarization on the surface of the platter.

During reads and writes, the R/W head senses the magnetic polarization and never touches the surface of the platter.

When the spindle is rotating, there is a microscopic air gap between the R/W heads and the platters, known as the head flying height.

This air gap is removed when the spindle stops rotating and the R/W head rests on a special area on the platter near the spindle.

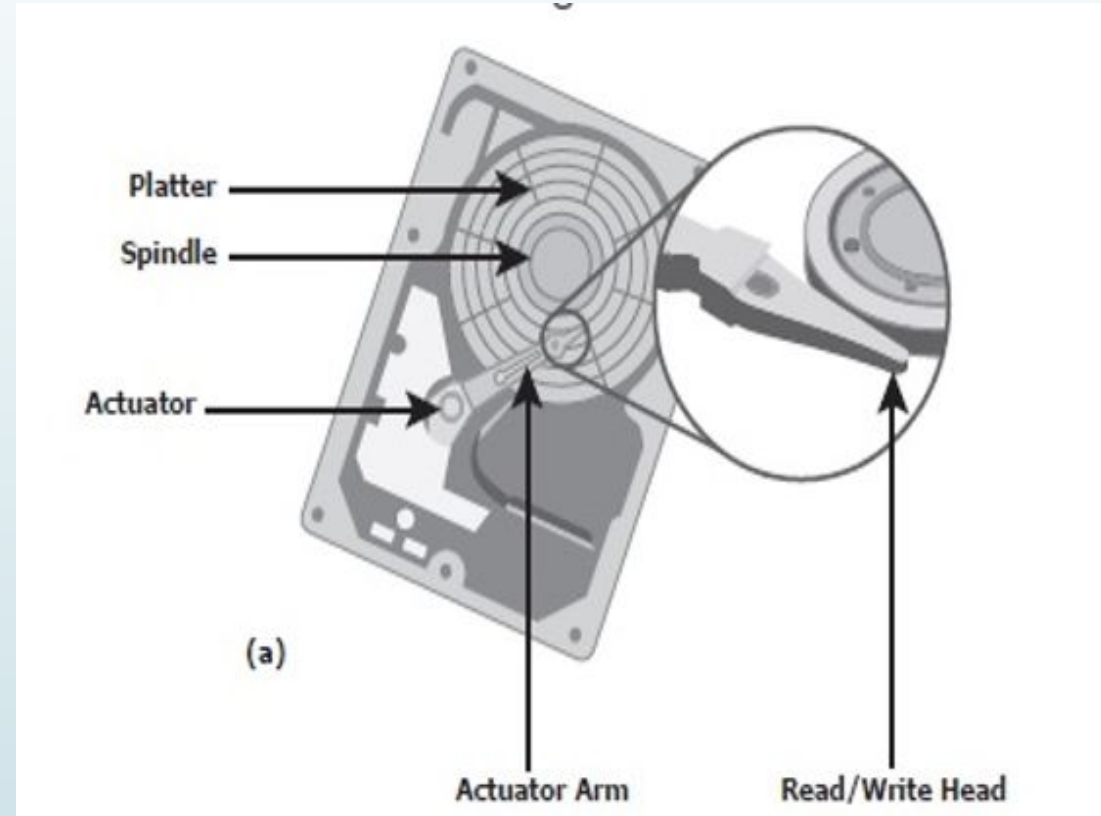
This area is called the landing zone. The landing zone is coated with a lubricant to reduce friction between the head and the platter.

Actuator Arm Assembly

The R/W heads are mounted on the actuator arm assembly (refer to Figure 2-2 [a]), which positions the R/W head at the location on the platter where the data needs to be written or read.

The R/W heads for all platters on a drive are attached to one actuator arm assembly and move across the platters simultaneously.

Note that there are two R/W heads per platter, one for each surface, as shown in Figure 2-4.



Controller

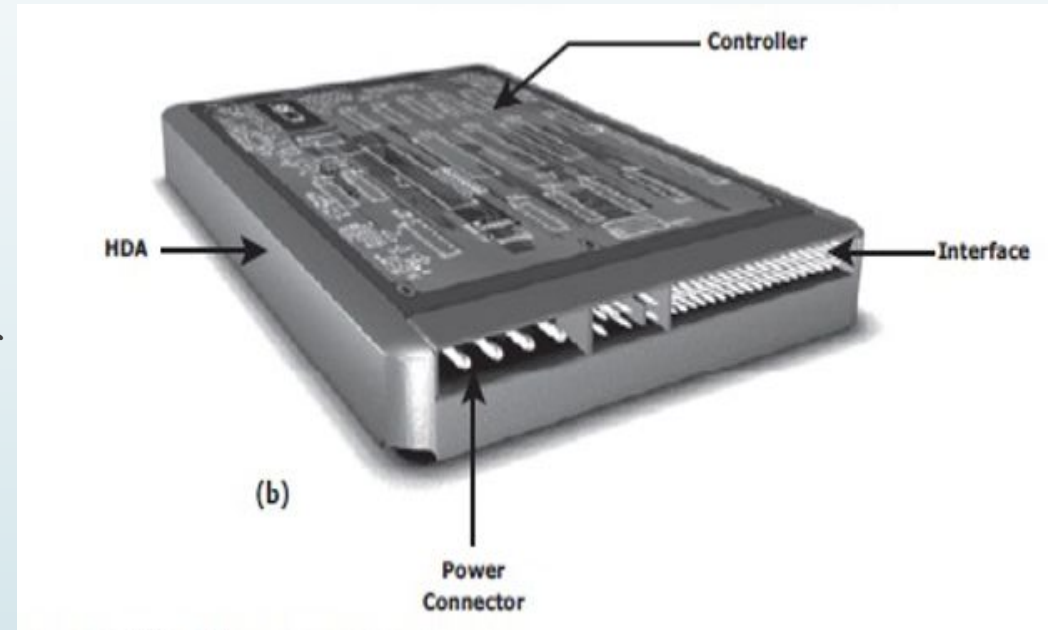
The controller (see Figure 2-2 [b]) is a printed circuit board, mounted at the bottom of a disk drive.

It consists of a microprocessor, internal memory, circuitry, and firmware.

The firmware controls power to the spindle motor and the speed of the motor.

It also manages communication between the drive and the host.

In addition, it controls the R/W operations by moving the actuator arm and switching between different R/W heads, and performs the optimization of data access.



Physical Disk Structure

Data on the disk is recorded on tracks, which are concentric rings on the platter around the spindle, as shown in Figure 2-5.

The tracks are numbered, starting from zero, from the outer edge of the platter.

The number of tracks per inch (TPI) on the platter (or the track density) measures how tightly the tracks are packed on a platter.

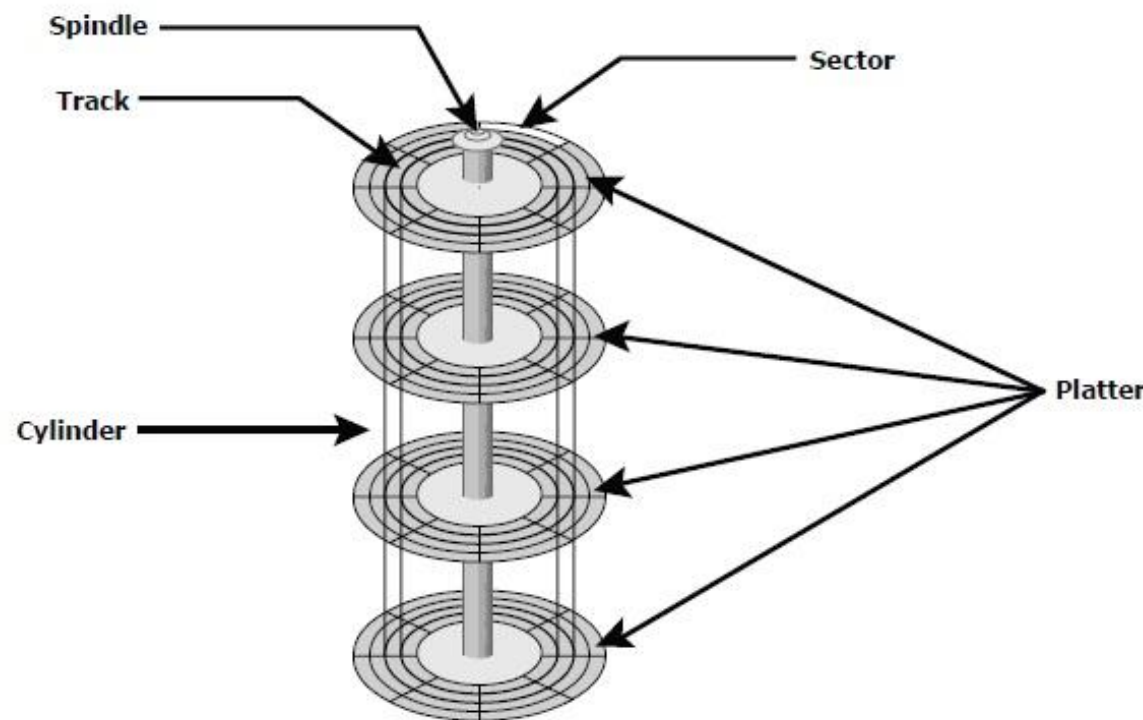


Figure 2-5: Disk structure: sectors, tracks, and cylinders

Logical Block Addressing

Earlier drives used physical addresses consisting of the cylinder, head, and sector (CHS) number to refer to specific locations on the disk, as shown in Figure 2-7 (a), and the host operating system had to be aware of the geometry of each disk being used.

Logical block addressing (LBA), shown in Figure 2-7 (b), simplifies addressing by using a linear address to access physical blocks of data.

The disk controller translates LBA to a CHS address, and the host only needs to know the size of the disk drive in terms of the number of blocks.

The logical blocks are mapped to physical sectors on a 1:1 basis.

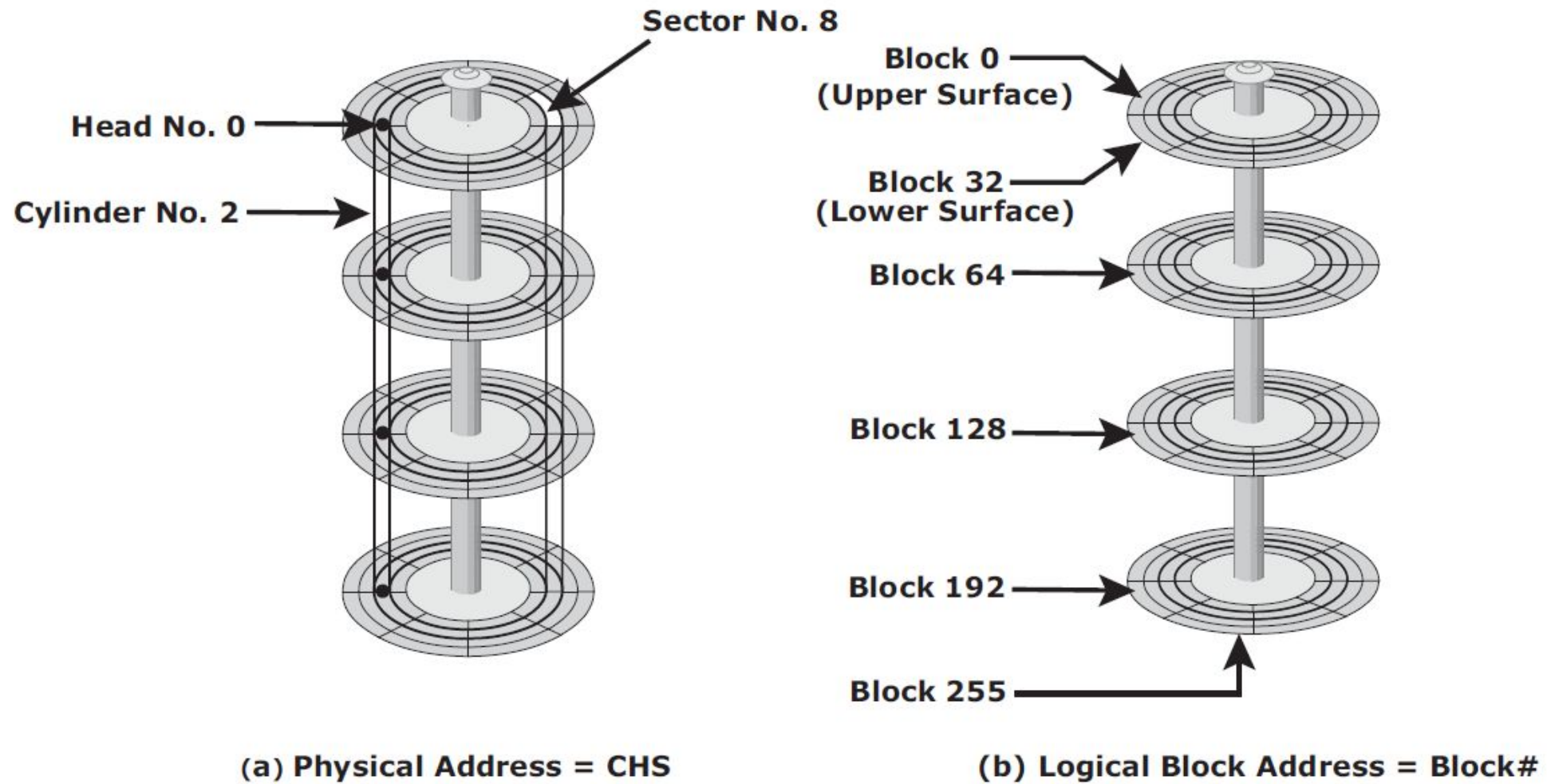


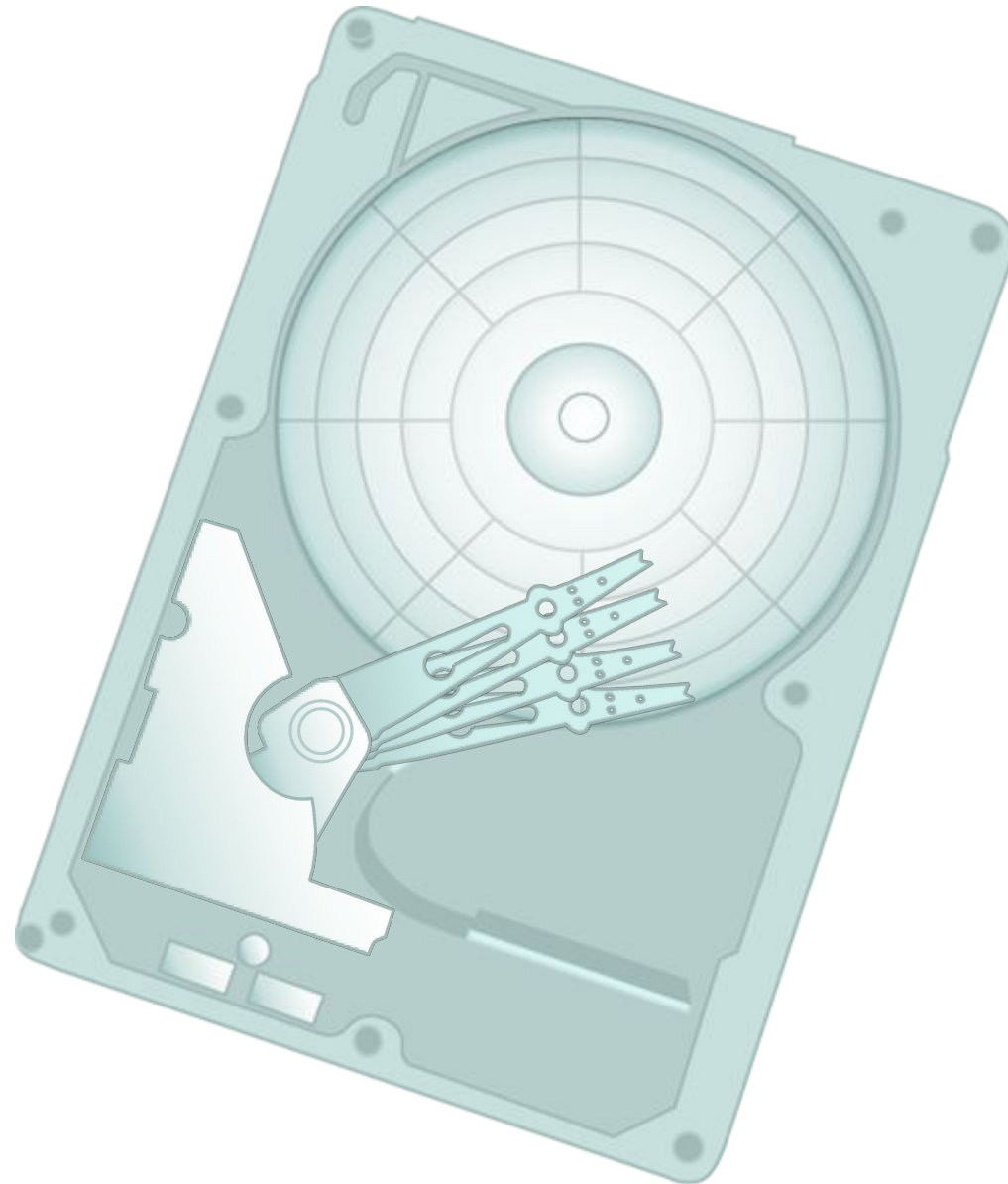
Figure 2-10: Physical address and logical block address

Disk Drive Performance

- Electromechanical device
 - Impacts the overall performance of the storage system
- Disk Service Time
 - Time taken by a disk to complete an I/O request
 - ▣ **Seek Time** - Full Stroke, Average, Track-to-Track
 - ▣ **Rotational Latency**
 - ▣ **Data Transfer Rate**

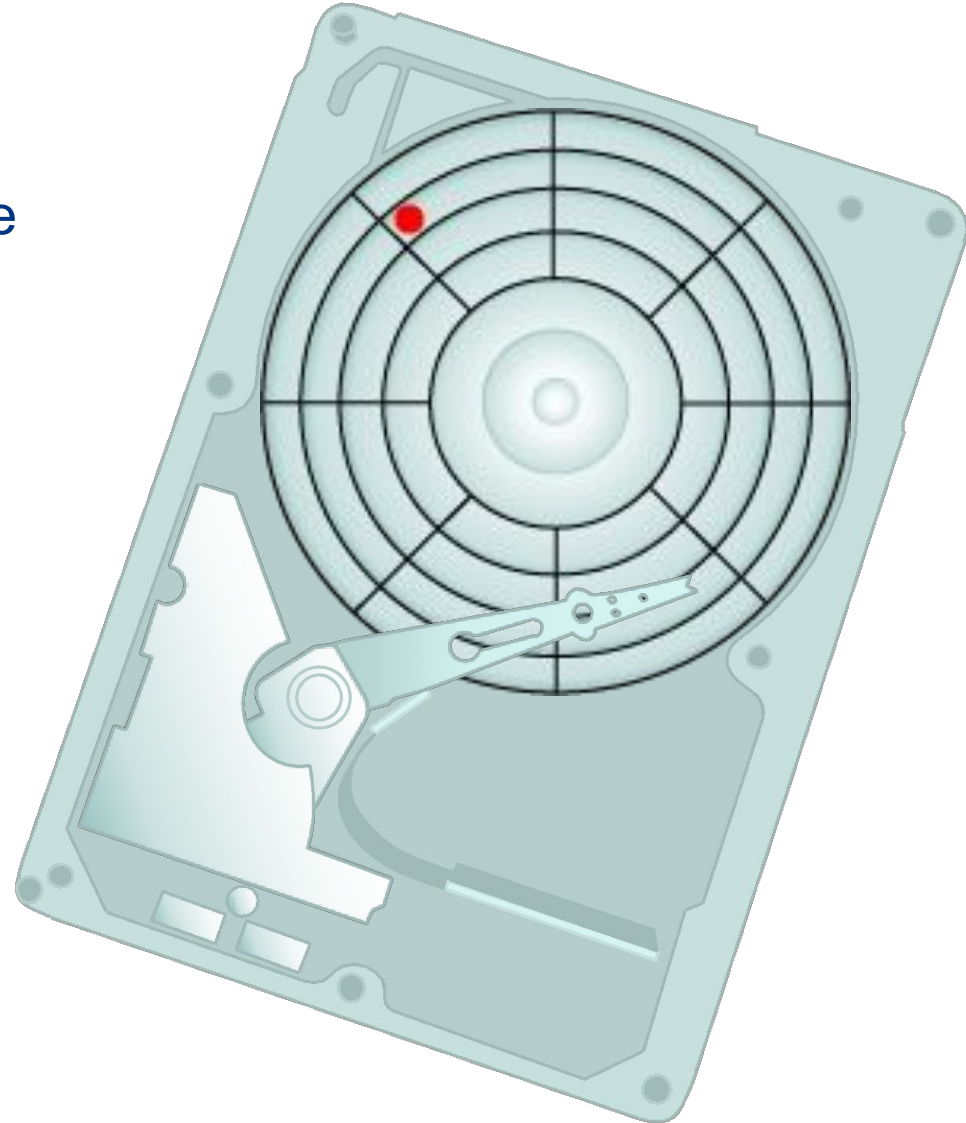
Disk Drive Performance: Seek Time

- **Seek Time** - Time taken to position the read/write head
- Lower the seek time, the faster the I/O operation
- Seek time specifications include:
 - **Full stroke**- The time taken by the R/W head to move across the entire width of the disk, from the innermost track to the outermost track.
 - **Average** - The average time taken by the R/W head to move from one random track to another, normally listed as the time for one-third of a full stroke.
 - **Track-to-track** - The time taken by the R/W head to move between adjacent tracks.
 - The average seek time on a modern disk is typically in the range of **3 to 15 milliseconds**.



Disk Drive Performance: Rotational Speed/Latency

- The time taken by platter to rotate and position the data under the R/W head
- Depends on the rotation speed of the spindle
- Average rotational latency
 - One-half of the time taken for a full rotation
 - Appx. 5.5 ms for 5400-rpm drive
 - Appx. 2.0 ms for 15000-rpm drive



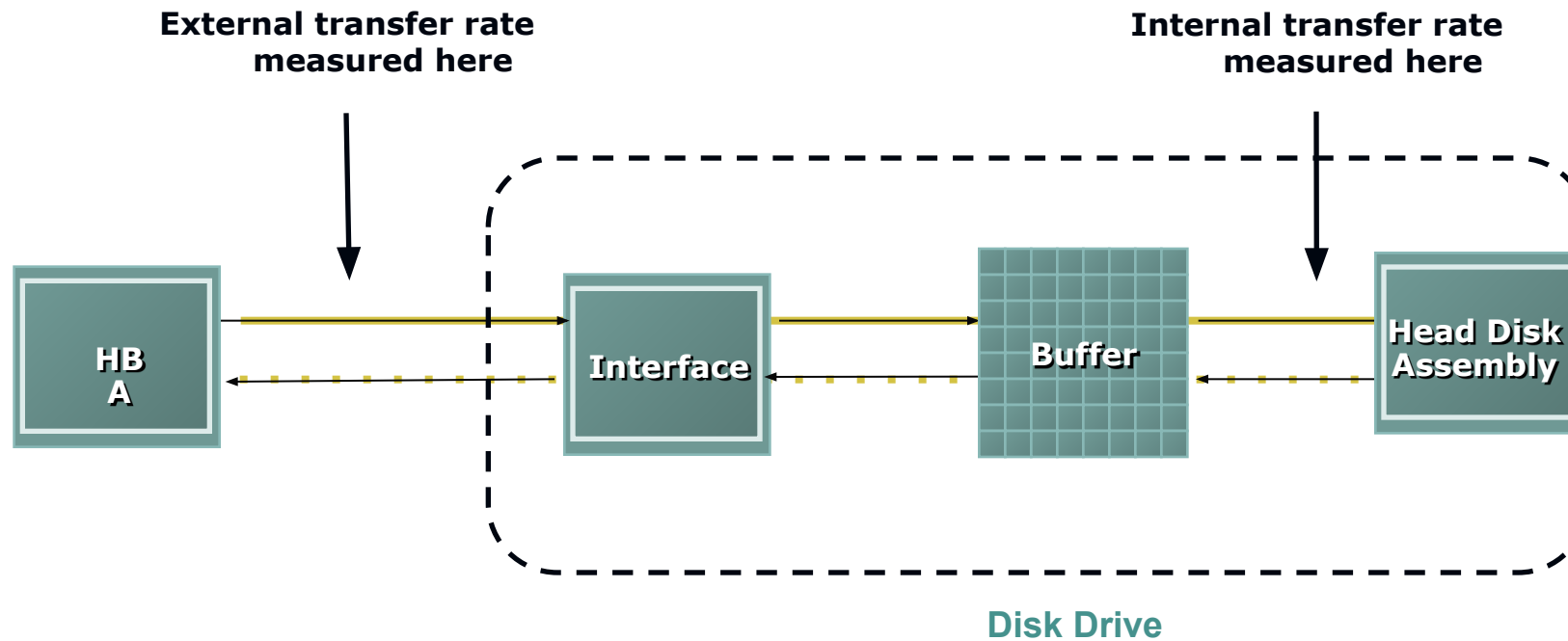
Rotational Latency

- The time taken by the platter to rotate and position the data under the R/W head is called rotational latency.
- This latency depends on the rotation speed of the spindle and is measured in milliseconds. The average rotational latency is one-half of the time taken for a full rotation.
- Average rotational latency is approximately 5.5 ms for a 5,400-rpm drive, and around 2.0 ms for a 15,000-rpm (or 250-rps revolution per second) drive as shown here:
- Average rotational latency for a 15,000 rpm (RPS=15,000/60=250 révolutions per second or 250 rps) drive = $0.5/250 = 2$ milliseconds

$$\text{Average Rotational Latency} = 1/2 \times \text{Time per Revolution}$$

Disk Drive Performance: Data Transfer Rate

- Average amount of data per unit time
- Internal Transfer Rate
 - Speed at which data moves from a track to disk internal buffer
- External Transfer Rate
 - The advertised speed of the interface



Power Consumption

SATA vs SAS drives in 3.5-in form factor in 2006

	Capacity(G B)	Price	Platters	RPM	Diameter(i nches)	Average seek (ms)	Power (watts)	I/O/sec	Disk BW (MB/sec)
SATA	500	\$375	4 or 5	7,200	3.7	8-9	12	117	31-65
SAS	37	\$150	1	15,000	2.6	3-4	25	285	85-142

Power consumed by the disk motor:

Power $\approx$ Diameter^{4.6} $\times$ RPM^{2.8} $\times$ Number of platters

Disk I/O Controller Utilization

To understand the laws of disk performance, a disk can be viewed as a black box consisting of two elements:

- **Queue:** The location where an I/O request waits before it is processed by the I/O controller.
- **Disk I/O Controller:** Processes I/O s that are waiting in the queue one by one.

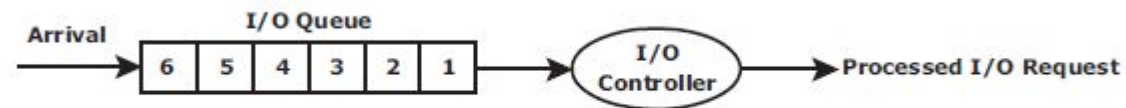


Figure 2-12: I/O processing

Fundamental Laws Governing Disk Performance

- Little's Law

- Describes the relationship between the number of requests in a queue and the response time.

- $N = a \times R$

- “N” is the total number of requests in the system

- “a” is the arrival rate

- “R” is the average response time

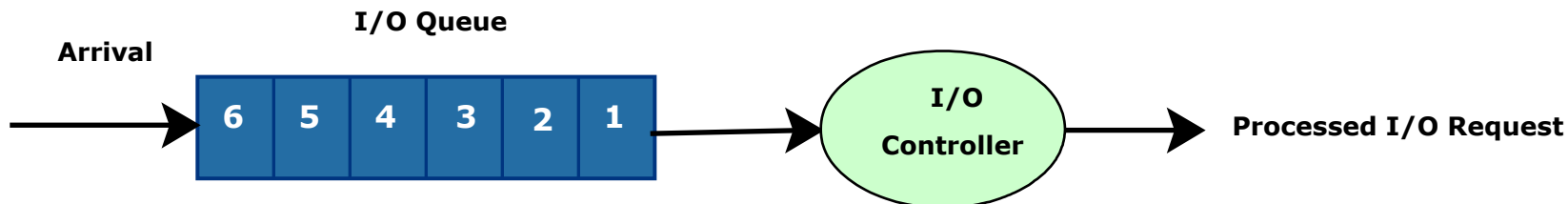
- Utilization law

- Defines the I/O controller utilization

- $U = a \times R_s$

- “U” is the I/O controller utilization

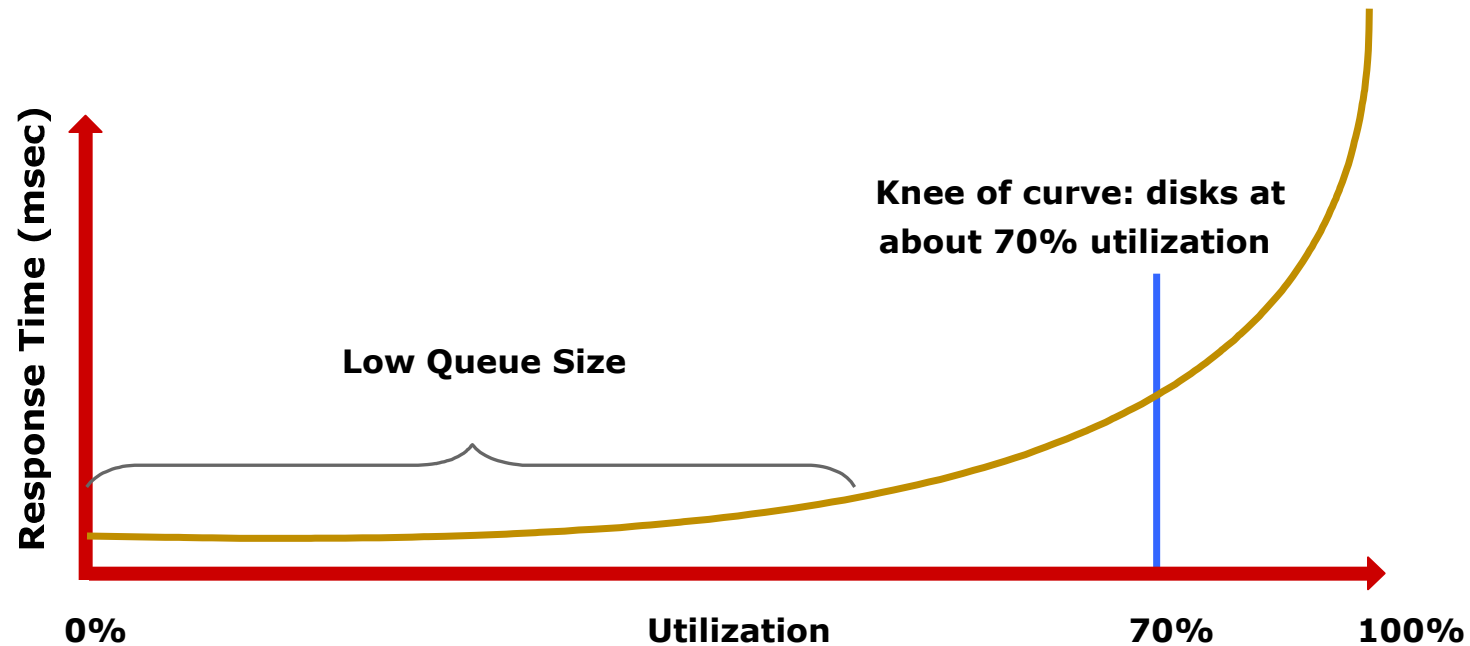
- “ R_s ” is the *service time*



Disk I/O Controller Utilization

- It can be concluded that by reducing the service time (the sum of seek time, latency, and internal transfer rate) or utilization by half, the response time can be reduced drastically (almost six times in the preceding example).
- Response time changes are nonlinear as utilization increases.
- When the average queue sizes are low, response time remains low.
- Response time increases slowly with added load on the queue, and increases exponentially when utilization exceeds 70 percent.

Utilization vs. Response time



- Consider a disk I/O system in which an I/O request arrives at a rate of 100 I/Os per second. The service time, R_s , is 4 ms.
 - Utilization of I/O controller ($U = a \times R_s$)
 - Total response time ($R = R_s / (1 - U)$)
 - Time spent by a request in a queue $R_q = R_s \cdot U / (1 - U)$
 - Average queue size = $U^2 / (1 - U)$
- Calculate the same with service time is doubled

Storage Design Based on Application Requirements and Disk Performance

62

- Determining storage requirements for an application is estimated by the size and number of file systems and database components used by applications.
- The I/O size, I/O characteristics, and the number of I/Os generated by the application at peak workload are other factors that affect disk performance, I/O response time, and design of storage systems.
- The I/O block size depends on the file system and the database on which the application is built.
- The disk service time **(TS)** for an I/O is a key measure of disk performance; TS, along with disk utilization rate **(U)**, determines the I/O response time for an application.
- The total disk service time **(TS)** is the sum of the seek time **(T)**, rotational latency **(L)**, and internal transfer time **(X)**: **$TS = T + L + X$**

Storage Design Based on Application Requirements and Disk Performance

Consider an example with the following specifications provided for a disk:

- The average seek time is 5 ms in a random I/O environment; therefore,
- $T = 5 \text{ ms}$.
- Disk rotation speed of 15,000 revolutions per minute or 250 revolutions per second — from which rotational latency (L) can be determined, which is one-half of the time taken for a full rotation or **Rotational latency (L): $L = (0.5/250 \text{ rps expressed in ms})$** .
- 40 MB/s internal data transfer rate, from which the internal transfer time
- **Internal transfer time (X):** is derived based on the block size of the I/O — for example, an I/O with a block size of 32 KB; therefore **$X = 32 \text{ KB}/40 \text{ MB}$** .
- Consequently, the time taken by the I/O controller to serve an I/O of block size 32 KB is
- $T_s = T + L + X$
- **Total Disk Service time, $T_s = 5 \text{ ms} + (0.5/250) + 32 \text{ KB}/40 \text{ MB} = 7.8 \text{ ms}$** .

Therefore, the maximum number of I/Os serviced per second or IOPS is

- **$(1/T_s) = 1/(7.8 \times 10^{-3}) = 128 \text{ IOPS}$** .

Storage Design Based on Application Requirements and Disk Performance

- The application response time, R , increases with an increase in disk controller utilization.
- For example, the response time (R) for an I/O with a block size of 32 KB at 96 percent disk controller utilization is
- $R = TS/(1 - U) = 7.8/(1 - 0.96) = 195 \text{ ms}$
- If the application demands a faster response time, then the utilization for the disks should be maintained below 70 percent.
- For the same 32-KB block size, at 70-percent disk utilization, the response time reduces drastically to **26 ms**.
- However, at lower disk utilization, the number of IOPS a disk can perform is also reduced.
- In the case of a 32-KB block size, a disk can perform 128 IOPS at almost 100 percent utilization, whereas the number of IOPS it can perform at 70-percent utilization is **89 (128 x 0.7)**.
- This indicates that the number of I/Os a disk can perform is an important factor that needs to be considered while designing the storage requirement for an application.
- Therefore, the storage requirement for an application is determined in terms of both the capacity and IOPS.
- If an application needs 200 GB of disk space, then this capacity can be provided simply with a single disk.
- However, if the application IOPS requirement is high, then it results in performance degradation because just a single disk might not provide the required response time for I/O operations.
- The total number of disks required (DR) for an application is computed as follows:
$$DR = \text{Max}(DC, DI)$$

Storage Design Based on Application Requirements and Disk Performance

- The total number of disks required (DR) for an application is computed as follows: $DR = \text{Max}(DC, DI)$
- Where DC is the number of disks required to meet the capacity, and DI is the number of disks required to meet the application IOPS requirement.
- **Eg:** Consider an example in which the capacity requirement for an application is 1.46 TB. The number of IOPS generated by the application at peak workload is estimated at 9,000 IOPS. The vendor specifies that a 146-GB, 15,000-rpm drive is capable of doing a maximum 180 IOPS.
- In this example, the number of disks required to meet the capacity requirements will be $1.46 \text{ TB} / 146 \text{ GB} = 10 \text{ disks}$.
- To meet the application IOPS requirements, the number of disks required is $9,000 / 180 = 50$.
- However, if the application is response-time sensitive, the number of IOPS a disk drive can perform should be calculated based on 70- percent disk utilization.
- Considering this, the number of IOPS a disk can perform at 70 percent utilization is $180 \times 0.7 = 126 \text{ IOPS}$.
- Therefore, the number of disks required to meet the application IOPS requirement will be $9,000 / 126 = 72$.
- As a result, the number of disks required to meet the application requirements will be $\text{Max}(10, 72) = 72 \text{ disks}$.
- The preceding example indicates that from a capacity-perspective, 10 disks are sufficient; however, the number of disks required to meet application performance is 72.
- Examples of these solutions are disk native command queuing, use of flash drives, RAID, and the use of cache memory.

Application Requirements and Disk Performance

Exercise:

- ❑ Consider an application that requires 1TB of storage capacity and performs 4900 IOPS
 - ❑ Application I/O size is 4KB
 - ❑ As it is business critical application, response time must be within acceptable range
- ❑ Specification of available disk drive:
 - ❑ Drive capacity = 73 GB
 - ❑ 15000 RPM
 - ❑ 5 ms average seek time
 - ❑ 40 MB/sec transfer rate

Calculate the number of disks required?

Solution

- Calculate time required to perform one I/O
Seek time + (rotational delay)/speed in RPM + (block size/transfer rate)
Therefore, $5 \text{ ms} + 0.5 / 15000 + 4\text{K} / 40\text{MB} = 7.1 \text{ msec}$
- Calculate max. number of IOPS a disk can perform
 - $1 / 7.1 \text{ ms} = 140 \text{ IOPS}$
- For acceptable response time disk controller utilization must be less than 70%
 - Therefore, $140 \times 0.7 = 98 \text{ IOPS}$
- To meet application
 - Performance requirement we need $4900 / 98$ i.e. 50 disk
 - Capacity requirement we need $1\text{TB} / 73 \text{ GB}$ i.e. 14 disk

Disk required = max (capacity, performance)

Storage Design Based on Application Requirements and Disk Performance

- The application response time, R , increases with an increase in disk controller utilization.
- For example, the response time (R) for an I/O with a block size of 32 KB at 96 percent disk controller utilization is
- **$R = TS/(1 - U) = 7.8/(1 - 0.96) = 195 \text{ ms}$**
- If the application demands a faster response time, then the utilization for the disks should be maintained below 70 percent.
- For the same 32-KB block size, at 70-percent disk utilization, the response time reduces drastically to **26 ms**.
- However, at lower disk utilization, the number of IOPS a disk can perform is also reduced.
- In the case of a 32-KB block size, a disk can perform 128 IOPS at almost 100 percent utilization, whereas the number of IOPS it can perform at 70-percent utilization is **89 (128 x 0.7)**.
- This indicates that the number of I/Os a disk can perform is an important factor that needs to be considered while designing the storage requirement for an application.
- Therefore, the storage requirement for an application is **determined in terms of both the capacity and IOPS**.
- If an application needs 200 GB of disk space, then this capacity can be provided simply with a single disk.
- However, if the application IOPS requirement is high, then it results in performance degradation because just a single disk might not provide the required response time for I/O operations.

Enterprise Flash Drives: A New Generation Drives

Conventional disk drive

- ❑ Mechanical Delay associated with conventional drive
 - ❑ Seek time
 - ❑ Rotational latency
- ❑ More power consumption due to mechanical operations
- ❑ Low Mean Time Between Failure



Enterprise flash drive

- ❑ Highest possible throughput per drive
 - ❑ No Spinning magnetic media
 - ❑ No Mechanical movement which causes seek and latency
 - ❑ Solid State enables consistent I/O performance
- ❑ Very low latency per I/O
- ❑ Energy efficient storage design
 - ❑ Lower power requirement per GB of storage
 - ❑ Lower power requirement per IOPS

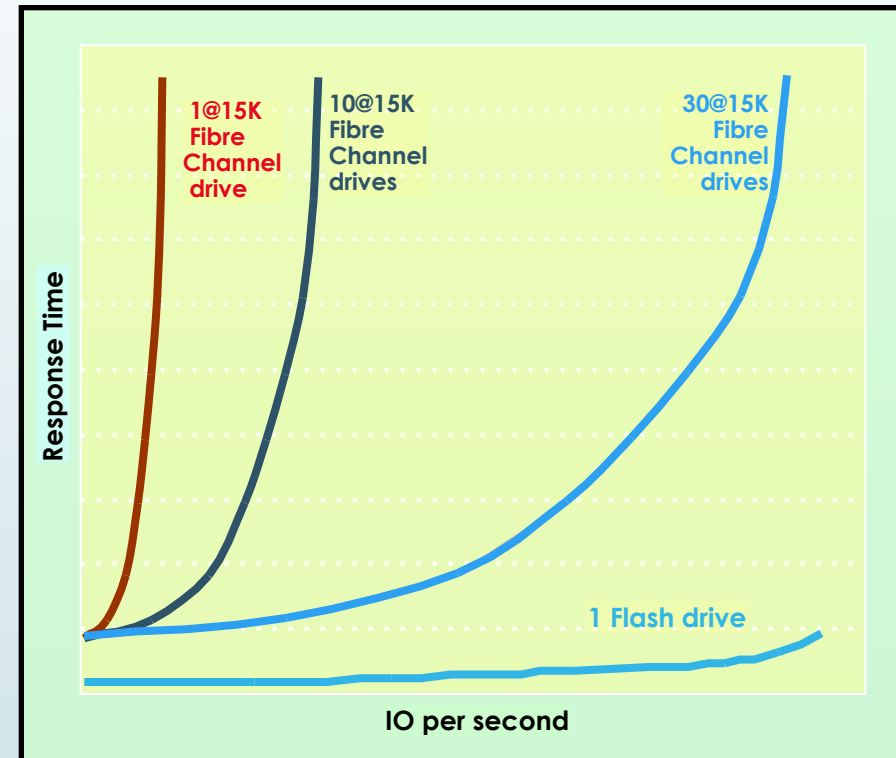
Enterprise Flash Drives – Overview

- Drive is based on Flash Solid State memory technology
 - High performance and low latency
 - Non volatile memory
 - Uses single layer cell (SLC) or Multi Level cell (MLC) to store data
- Enterprise Flash Drives use a 4GB FC interface



Enterprise Flash Drives – Benefits

- Faster performance
 - Up to 30 times greater IOPS (benchmarked)
 - Typical applications: 8 – 12X
 - Less than 1 millisecond service time
- More energy efficient
 - 38 percent less per terabyte
 - 98 percent less per IO
- Better reliability
 - No moving parts
 - Faster RAID rebuilds



Enterprise Flash Drives – “Tier-0” Application

- Position Enterprise Flash Drives as the high-performance option in demanding environments
 - Low latency applications, also known as “Tier-0” applications
- Standard form-factor and capacity design allows for easier integration
- High performance, low power for a “Green” initiative
- Target Customer/Market Segments:
 - High performance solutions coupled with low power
 - Specifically target Oracle database customers initially
 - Financial trading
 - OLTP databases

Concept in Practice: VMware ESXi

VMware is the leading provider for a server virtualization solution.

VMware ESXi provides a platform called hypervisor.

The hypervisor abstracts CPU, memory, and storage resources to run multiple virtual machines concurrently on the same physical server.

VMware ESXi is a hypervisor that installs on x86 hardware to enable server virtualization. It enables creating multiple virtual machines (VMs) that can run simultaneously on the same physical machine.

A VM is a discrete set of files that can be moved, copied, and used as a template. All the files that make up a VM are typically stored in a single directory on a cluster file system called Virtual Machine File System (VMFS).

The physical machine that houses ESXi is called the ESXi host. ESXi hosts provide physical resources used to run virtual machines. ESXi has two key components: VMkernel and Virtual Machine Monitor.

Concept in Practice: VMware ESXi

VMkernel provides functionality similar to that found in other operating systems, such as process creation, file system management, and process scheduling.

It is designed to specifically support running multiple VMs and provide core functionality such as resource scheduling, I/O stacks, and so on.

The virtual machine monitor is responsible for executing commands on the CPUs and performing Binary Translation (BT).

A virtual machine monitor performs hardware abstraction to appear as a physical machine with its own CPU, memory, and I/O devices.

Each VM is assigned a virtual machine monitor that has a share of the CPU, memory, and I/O devices to successfully run the VM.

Logical Components of the Host

The logical components of a host consist of the software applications and protocols that enable data communication with the user as well as the physical components. Following are the logical components of a host:

- ❑ **Operating system**
- ❑ **Device drivers**
- ❑ **Volume manager**
- ❑ **File system**
- ❑ **Application**

Logical Components of the Host

- *Application*

- Interface between user and the host
- Three-tiered architecture
 - Application UI, computing logic and underlying databases
- Application data access can be classified as:
 - **Block-level access:** Data stored and retrieved in blocks, specifying the LBA
 - **File-level access:** Data stored and retrieved by specifying the name and path of files

- *Operating system*

- Resides between the applications and the hardware
- Controls the environment

Logical Components of the Host (Cont)

- Device Drivers

- Enables operating system to recognize the device
- Provides API to access and control devices
- Hardware dependent and operating system specific

- File System

- File is a collection of related records or data stored as a unit
- File system is hierarchical structure of files
 - Examples: FAT 32, NTFS, UNIX FS and EXT2/3

Volume Manager

Disk partitioning was introduced to improve the flexibility and utilization of HDDs. In partitioning, an HDD is divided into logical containers called logical volumes (LVs) (see Figure 2-11).

For example, a large physical drive can be partitioned into multiple LVs to maintain data according to the file system's and applications' requirements.

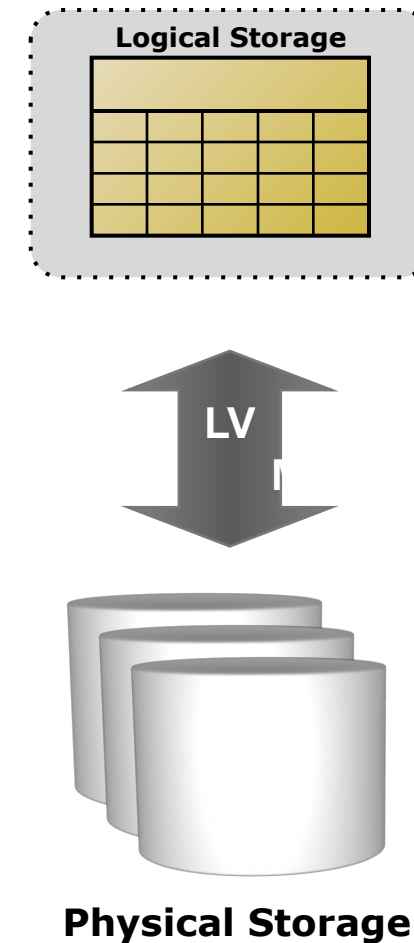
The partitions are created from groups of contiguous cylinders when the hard disk is initially set up on the host.

The host's file system accesses the partitions without any knowledge of partitioning and the physical structure of the disk.

Concatenation is the process of grouping several smaller physical drives and presenting them to the host as one logical drive (see Figure 2-11).

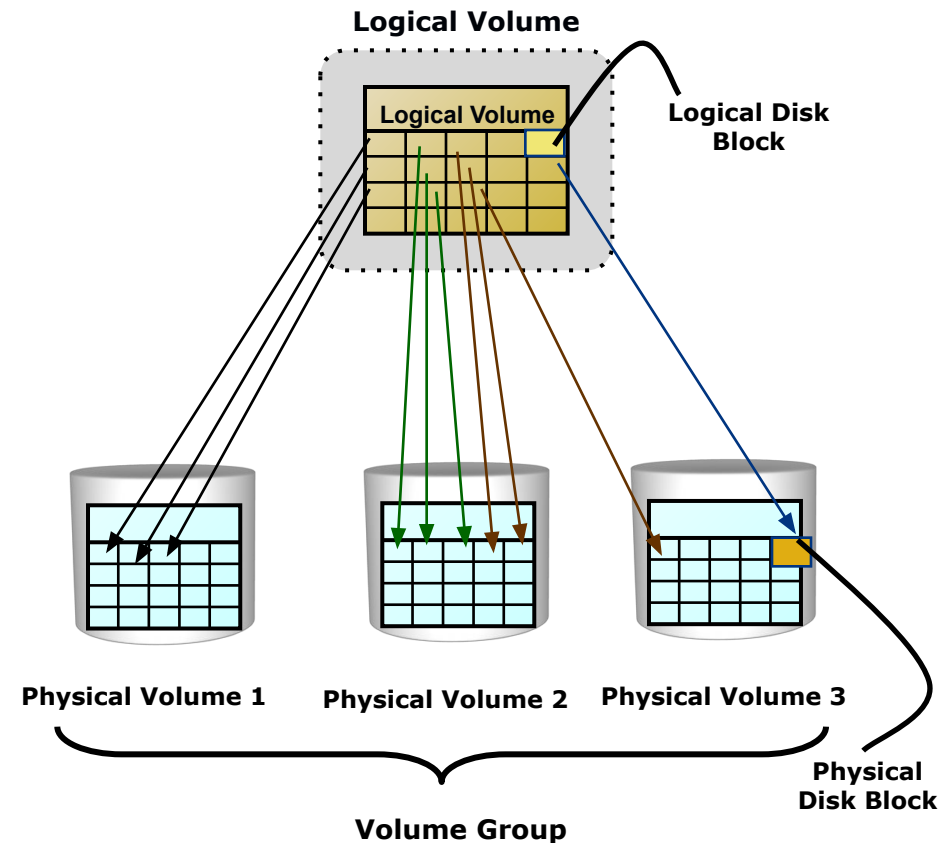
Logical Components of the Host: LVM

- Responsible for creating and controlling host level logical storage
 - Physical view of storage is converted to a logical view by mapping
 - Logical data blocks are mapped to physical data blocks
- Usually offered as part of the operating system or as third party host software
- LVM Components:
 - Physical Volumes
 - Volume Groups
 - Logical Volumes

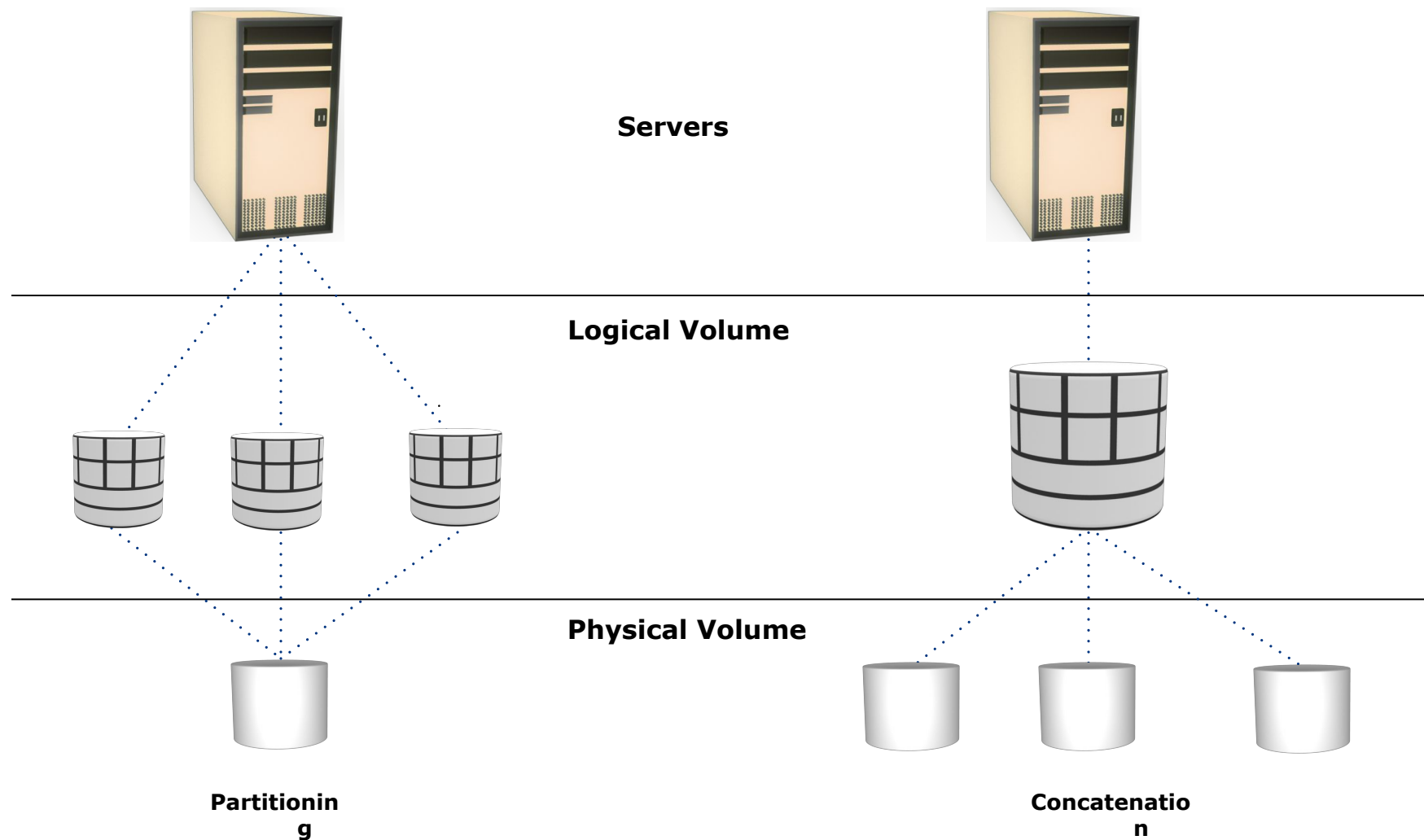


Volume Groups

- One or more Physical Volumes form a Volume Group
- LVM manages Volume Groups as a single entity
- Physical Volumes can be added and removed from a Volume Group as necessary
- Physical Volumes are typically divided into contiguous equal-sized disk blocks
- A host will always have at least one disk group for the Operating System
 - Application and Operating System data maintained in separate volume groups



LVM Example: Partitioning and Concatenation



LVM Example

A Windows XP system with two physical hard disks.

Physical Disk	Partition	Filesystem	Drive Letter
Hard Disk 1	Partition 1	NTFS	C:
	Partition 2	FAT32	D:
Hard Disk 2	Partition 1	FAT32	E:

- "C:", "D:", and "E:" are volumes.
- Hard Disk 1 and Hard Disk 2 are physical disks.
- Any of these can be called a "drive".

Operating System

An operating system controls all aspects of the computing environment.

It works between the application and physical components of the computer system.

One of the services it provides to the application is data access.

Device Driver

A device driver is special software that permits the operating system to interact with a specific device, such as a printer, a mouse, or a hard drive.

A device driver enables the operating system to recognize the device and to use a standard interface (provided as an application programming interface, or API) to access and control devices.

Device drivers are hardware dependent and operating system specific.

File System

A file is a collection of related records or data stored as a unit with a name.

A file system is a hierarchical structure of files.

File systems enable easy access to data files residing within a disk drive, a disk partition, or a logical volume.

A file system needs host-based logical structures and software routines that control access to files.

It provides users with the functionality to create, modify, delete, and access files.

Access to the files on the disks is controlled by the permissions given to the file by the owner, which are also maintained by the file system.

- ❑ **FAT 32 (File Allocation Table) for Microsoft Windows**
- ❑ **NT File System (NTFS) for Microsoft Windows**
- ❑ **UNIX File System (UFS) for UNIX**
- ❑ **Extended File System (EXT2/3) for Linux**

A file system block is the smallest “container” of physical disk space allocated for data.

Each file system block is a contiguous area on the physical disk.

The block size of a file system is fixed at the time of its creation.

File system size depends on block size and the total number of blocks of data stored.

A file can span multiple file system blocks because most files are larger than the predefined block size of the file system.

File system blocks cease to be contiguous (i.e., become fragmented) when new blocks are added or deleted.

Over time, as files grow larger, the file system becomes increasingly fragmented. Figure 2-12 shows the following process of mapping user files to the disk storage subsystem with an LVM:

File System: Metadata Examples

UNIX (UFS)

- File type and permissions
- Number of links
- Owner and group IDs
- Number of bytes in the file
- Last file access
- Last file modification

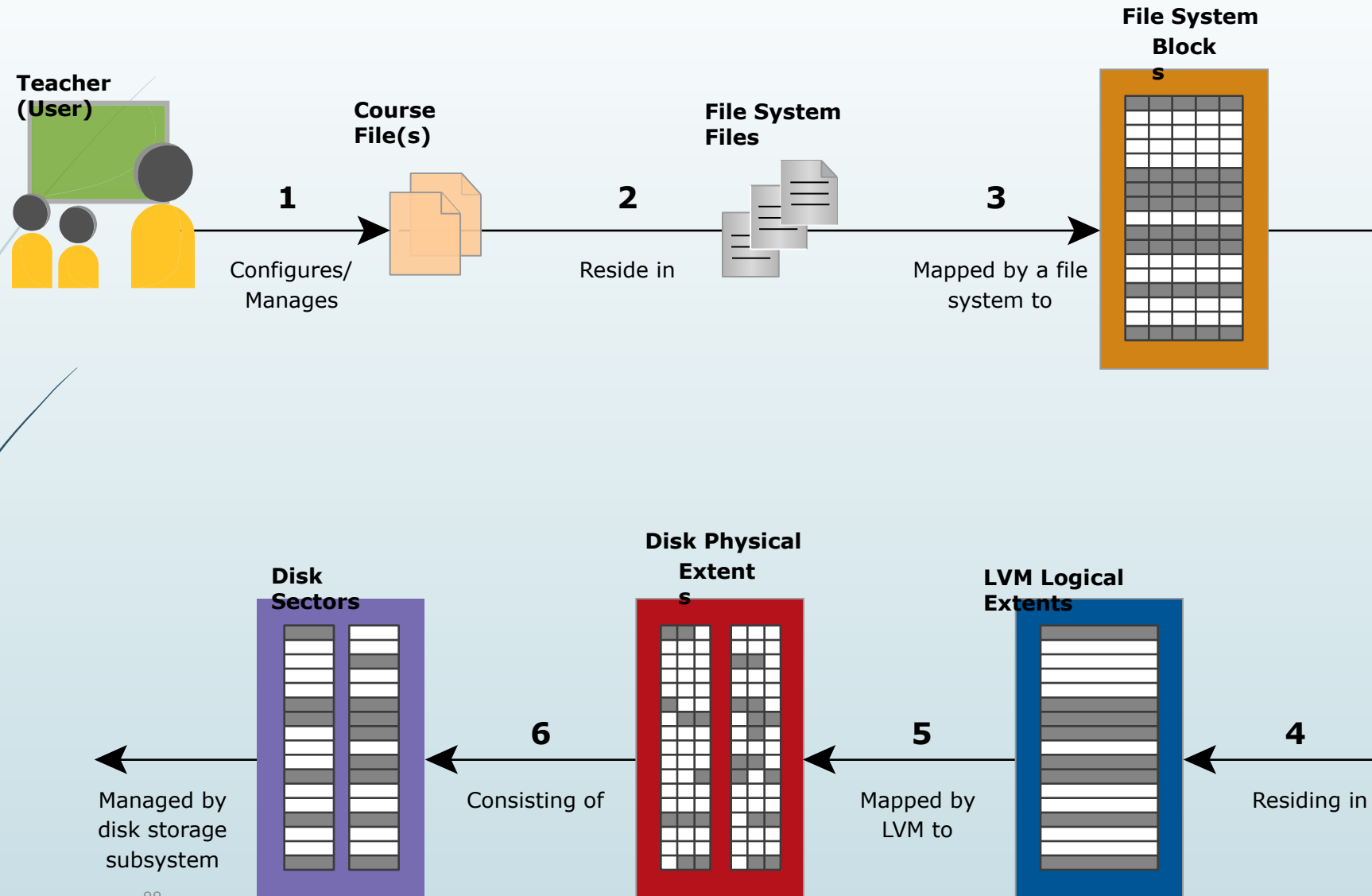
Windows (NTFS)

- Time stamp and link count
- File name
- Access rights
- File data
- Index information
- Volume information

File Systems: Journaling and Logging

- Improves data integrity and system restart time over non-journaling file systems.
- Uses a separate area called a log or journal.
 - May hold all data to be written
 - May hold only metadata
- Disadvantage - slower than other file systems.
 - Each file system update requires at least 1 extra write – to the log

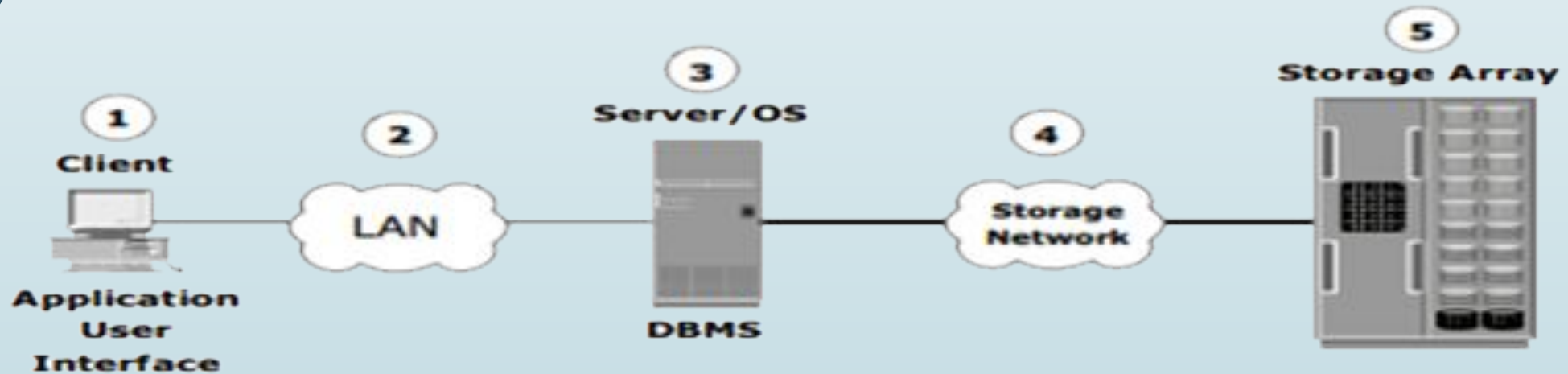
How Files are Moved to and from Storage



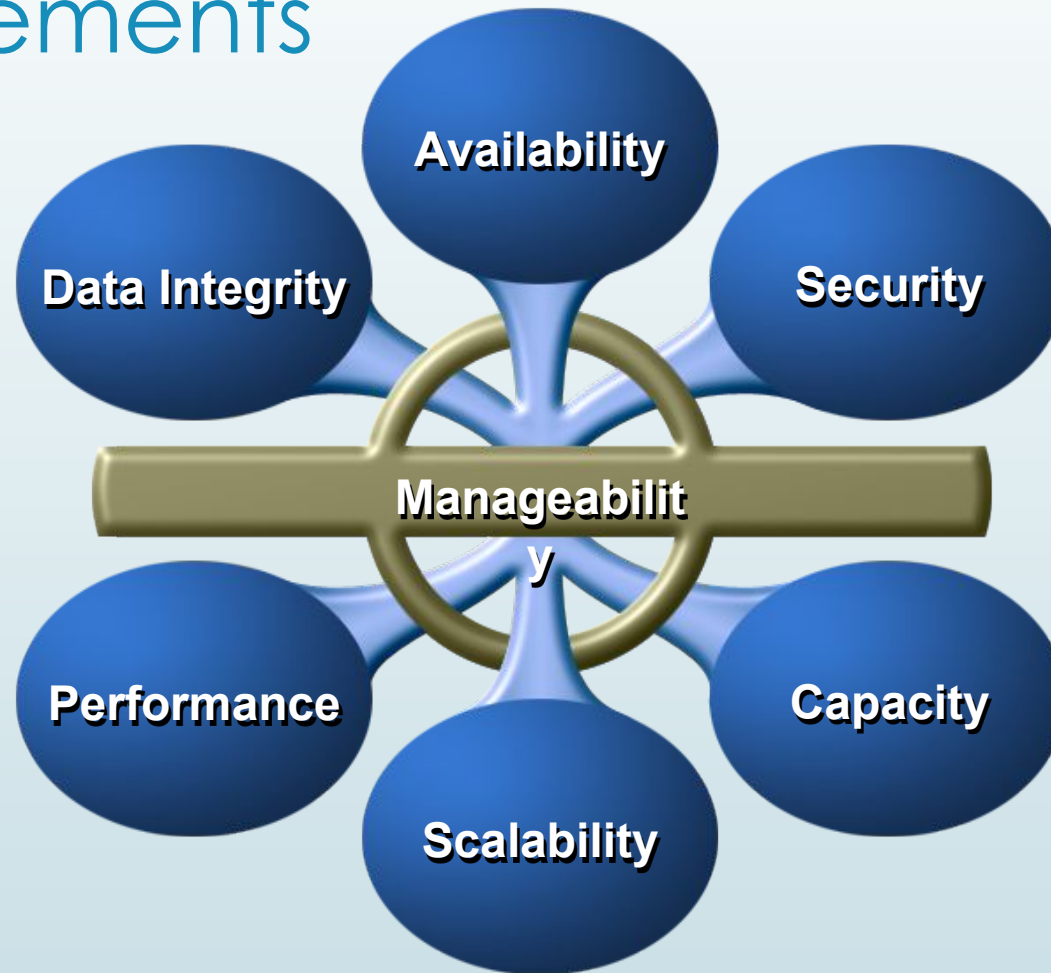
- 1) Files are created and managed by users and applications.
- 2) These files reside in the file systems.
- 3) The file systems are then mapped to units of data, or file system blocks.
- 4) The file system blocks are mapped to logical extents.
- 5) These in turn are mapped to disk physical extents either by the operating system or by the LVM.
- 6) These physical extents are mapped to the disk storage subsystem.

Core elements of a Data center

- ❑ **Application:** An application is a computer program that provides the logic for computing operations.
- ❑ **Database :** A Database is an organized collection of Data.
- ❑ **Server and operating system:** A computing platform that runs applications and databases.
- ❑ **Network:** A data path that facilitates communication between clients and servers or between servers and storage.
- ❑ **Storage array:** A device that stores data persistently for subsequent use.



Key Requirements for Data Center Elements



Key Requirements for Data Center Elements

- **Availability** : Accessibility should be insured for a business implementation.
- **Security** : Policies, procedures, and proper integration of the data center core elements that will prevent unauthorized access to information must be established.
- **Scalability** : Additional On-Demand services , without interrupting business operations.
- **Performance**: All the core elements of the data center should be able to provide optimal performance and service all processing requests at high speed.
- **Data integrity** : Mechanisms such as error correction codes and parity bits which ensure that data is written to disk exactly as it was received.
- **Capacity** : To increase and decrease core elements capacity on demand.
- **Manageability**: A data center should perform all operations and activities in the most efficient manner. Manageability can be achieved through automation and the reduction of human (manual) intervention in common tasks.

Evolution of Storage Technology and Architecture

- Direct-attached storage (DAS)
- Just a Bunch Of Disk(JBOD)
- Redundant Array of Independent Disks (RAID)
- Network-attached storage (NAS)
- Storage area network (SAN)
- Internet Protocol SAN (IP-SAN)

Information Lifecycle Management(ILM)

Information lifecycle management (ILM) is a proactive strategy that enables an IT organization to effectively manage the data throughout its lifecycle, based on predefined business policies.

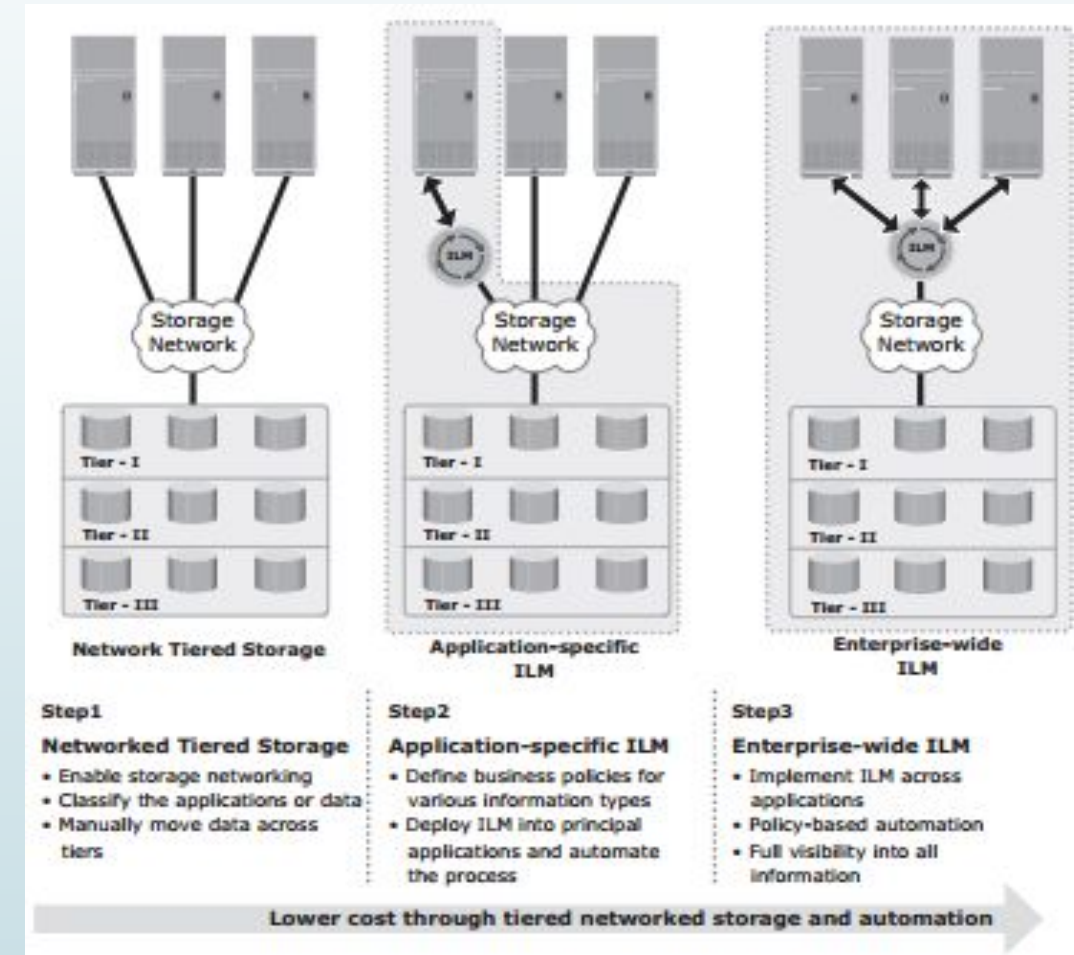
This allows an IT organization to optimize the storage infrastructure for maximum return on investment.

An ILM strategy should include the following characteristics:

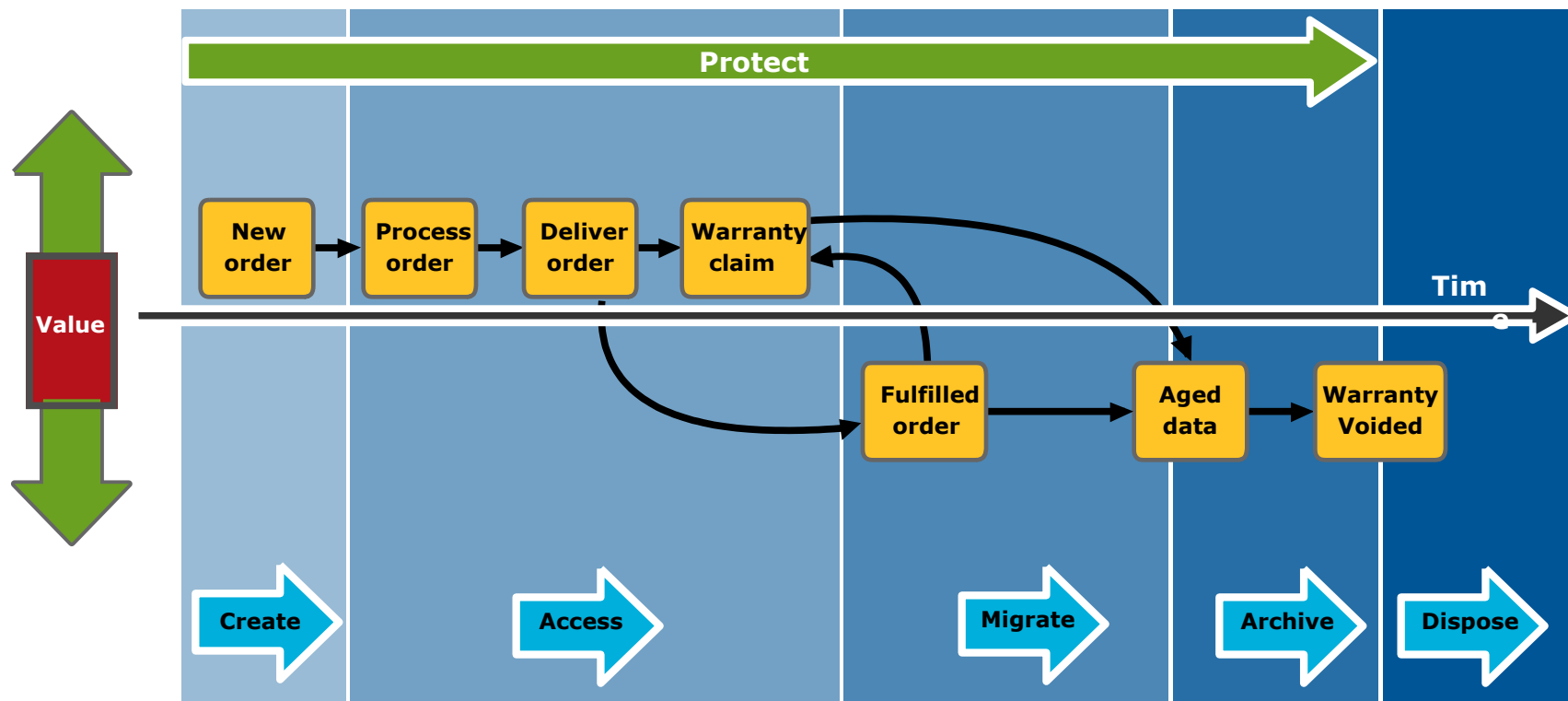
- ❑ Business-centric: It should be integrated with key processes, applications, and initiatives of the business to meet both current and future growth in information.
- ❑ Centrally managed: All the information assets of a business should be under the purview of the ILM strategy.
- ❑ Policy-based: The implementation of ILM should not be restricted to a few departments. ILM should be implemented as a policy and encompass all business applications, processes, and resources.
- ❑ Heterogeneous: An ILM strategy should take into account all types of storage platforms and operating systems.
- ❑ Optimized: Because the value of information varies, an ILM strategy should consider the different storage requirements and allocate storage resources based on the information's value to the business.

ILM Implementation

- Classifying data and applications on the basis of business rules and policies to enable differentiated treatment of information
- Implementing policies by using information management tools, starting from the creation of data and ending with its disposal
- Managing the environment by using integrated tools to reduce operational complexity
- Organizing storage resources in tiers to align the resources with data classes, and storing information in the right type of infrastructure based on the information's current value



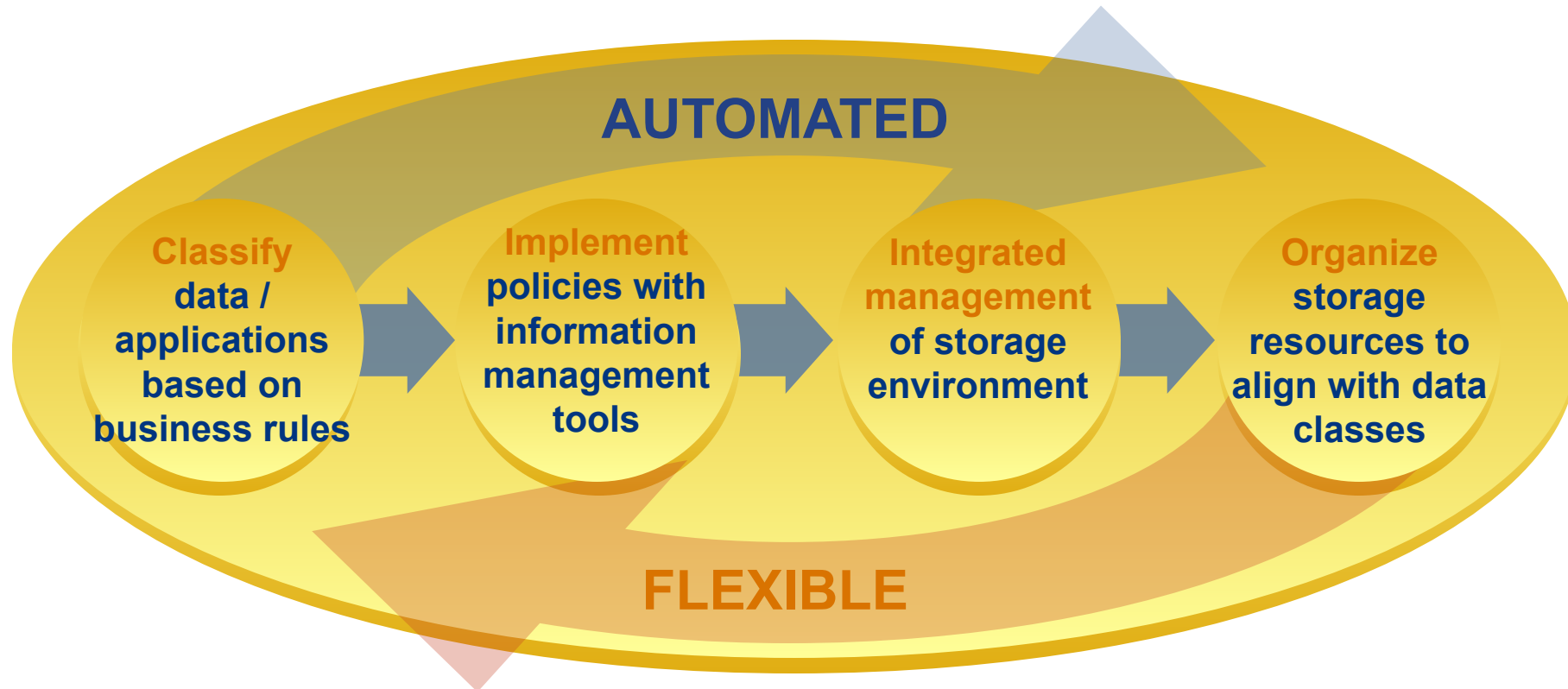
Information Lifecycle Management



A proactive strategy that enables an IT organization to effectively manage the data throughout its lifecycle

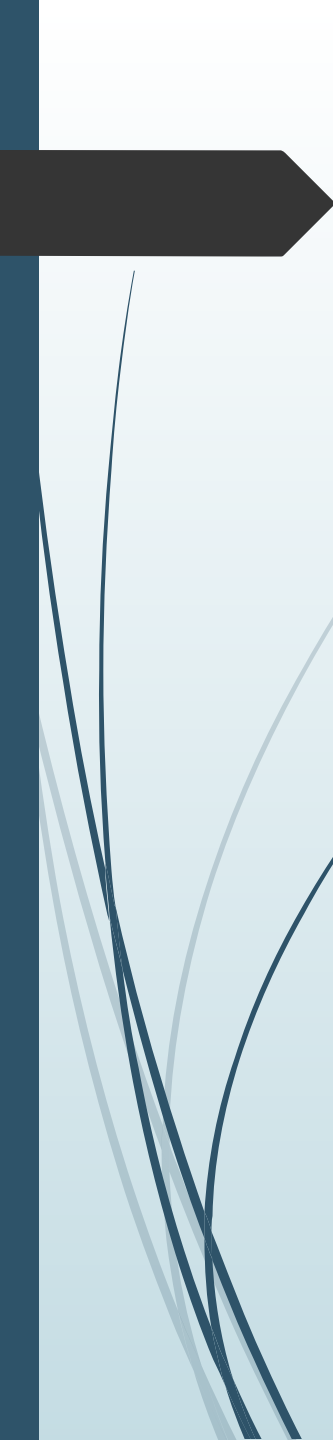
Information Lifecycle Management Process

Policy-based Alignment of Storage Infrastructure with Data Value



Benefits of Implementing ILM

- Improved utilization
 - Tiered storage platforms
- Simplified management
 - Processes, tools and automation
- Simplified backup and recovery
 - A wider range of options to balance the need for business continuity
- Maintaining compliance
 - Knowledge of what data needs to be protected for what length of time
- Lower Total Cost of Ownership
 - By aligning the infrastructure and management costs with information value



Key Challenges in Managing Information

- ❑ Exploding digital universe: The rate of information growth is increasing exponentially. Duplication of data to ensure high availability and repurposing has also contributed to the multifold increase of information growth.
- ❑ Increasing dependency on information: The strategic use of information plays an important role in determining the success of a business and provides competitive advantages in the marketplace.
- ❑ Changing value of information: Information that is valuable today may become less important tomorrow. The value of information often changes over time. Framing a policy to meet these challenges involves understanding the value of information over its lifecycle.



Summary

Key points covered in this lesson:

- ❑ Disk drive components and geometry
- ❑ Disk drive addressing scheme
- ❑ Disk drive performance
- ❑ Convention drive Vs Enterprise Flash Drives
- ❑ Enterprise Flash Drives for high performance and low power storage solution
- ❑ Storage system environment components:
 - ❑ Host, connectivity and storage
- ❑ Physical disk structure and addressing
- ❑ Factors affecting disk performance
- ❑ Flash drives benefits